

Public-Data Reproduction of GIGA-Lens Modeling for the DESI Strong Lens Foundry Demonstration System DESI-165.4754-06.0423 — Development Record*

Greg Benson[†]

Xiaosheng Huang[‡]

May 2026

commit: ef4c3c1

Phase 1 Internal Reproduction Report — full development record; final results in main.pdf

*This document is the complete development record (v1→v13, May–June 2026) of the Foundry-I reproduction, preserved as written. The concise final-state report — final posterior, corrected data treatment, and lessons for the GIGA-LENS stack — is **main.pdf** in this directory, served on the project site (<https://github.com/usfcs-edu/agentlc-lensing/blob/main/reproductions/foundry-i/papers/main.pdf>).*

Abstract

We present an end-to-end, public-data reproduction of the GIGA-Lens modeling demonstration for the DESI Strong Lens Foundry I system DESI-165.4754-06.0423, originally reported by Huang et al. (2025). Starting from the same *Hubble Space Telescope* WFC3/F140W exposure (GO-15867; MAST DOI 10.17909/hx0v-9260) and using only open-source infrastructure — GIGA-LENS, LENSTRONOMY, PHOTUTILS, JAX, and TENSORFLOW PROBABILITY — we rebuild the data products, construct an empirical PSF from in-field stars, and fit an EPL+external shear mass model together with four Sérsic lens-light components, a Sérsic source, and a shapelet basis of order $n_{\text{max}} = 6$ (74 free non-linear parameters). The naive `gigalens HMC()` helper did not JIT-compile at this model size under JAX 0.6.2; we trace this to a `pmap-over-all-tens-devices` wrapper, a 31-channel grouped `lstsq` convolution, and cuDNN autotuner thrashing, and show that with single-device execution, autotuning disabled, and a smooth marginalized target, genuine HMC compiles in ~ 35 s and samples the correct mode. For the headline result we nonetheless report the v10 Gaussian stochastic variational posterior (initialized from the best paper-mode chain of a 200-chain multi-start MAP search); a v11 NUTS run started from the v7 paper-mode MAP point confirms posterior multimodality and locates a third distinct local mode. A subsequent HMC investigation (§5.6) further corrects a saddle in the paper-mode MAP ($\Delta \log p \approx +62,000$), introduces an exact regularized Gaussian marginalization of the linear light amplitudes that fixes a non-smoothness in `gigalens lstsq_simulate`, and shows that the residual sampling difficulty is degeneracy-limited mixing ($\hat{R} > 1$ at 500 samples, $\text{cond} \sim 10^8$) rather than a sampler defect. A long 8-chain run (64,000 samples) then revises the slope picture: all chains

*All work reported here — the code, experiments, analysis, and text — was performed by Claude Code (Anthropic), running the Claude Opus 4.8 and Claude Fable 5 models, guided by Greg Benson.

[†]`benson@cs.usfca.edu`; Department of Computer Science, University of San Francisco, CA 94117, USA

[‡]`xhuang@usfca.edu`; Department of Physics & Astronomy, University of San Francisco, CA 94117; and Physics Division, Lawrence Berkeley National Laboratory, CA 94720

drift from the MAP/SVI $\gamma_{\text{EPL}} \approx 1.86$ *down through* the paper’s 1.372 toward $\gamma_{\text{EPL}} \approx 1.17$ at *higher* log-posterior, so the paper’s slope is consistent with — not far worse than — our model (an earlier $-32,500$ cold-start estimate was a stalled optimization), though the chains remain unconverged ($\hat{R} \leq 20$). Our 10 000-sample variational posterior recovers all six sign quadrants of the mass model, matches θ_{E} to 3.0%, e_1 to 2.5%, and the external shear position angle to within 1° of the published HMC result; the v11f NUTS mode reproduces θ_{E} to 1.6%, the closest of any version to date. The remaining gap — a 12° rotation of the lens ellipticity vector, $\sim 30\%$ smaller $|e_2|$ and $|\gamma_{\text{ext}}|$, and a 57% steeper deprojected slope $\gamma_{\text{EPL}} = 2.15$ versus the published 1.37 — is shown to be coherent rather than scattered. (An earlier draft attributed it to a missing model ingredient, “plausibly a mass–light position-angle tie or higher-order multipole correction”; that hypothesis is retracted — the published model contains neither, its Table 2 eccentricity priors being fully independent, and the campaign of §8 closed the gap through data treatment and a PSF sampling-convention fix alone.) We document upstream issues in GIGA-LENS 2.0/JAX 0.6.2/TFP 0.25.0 suitable for filing as GitHub issues, and provide the full software stack table to support future Foundry-series reproductions. A June 2026 paper-scale campaign on NERSC Perlmutter (§8), implementing the research group’s data-treatment feedback (tight crop, interloper masks, corrected noise) at the published compute scale, followed by a rigor revision (§8.5) that audited the noise calibration and uncovered a PSF sampling-convention defect in all native-scale fits, **retires the slope discrepancy**: with the convention-correct PSF the native-scale MAP floor drops from $\chi_\nu^2 \approx 3.4$ to 1.05 on identical noise, and the corrected HMC posterior gives $\gamma_{\text{EPL}} = 1.433 [1.400, 1.468]$ — consistent with the published 1.372 ± 0.023 at $< 2\sigma$, with a ~ 0.1 model-class systematic — $\theta_{\text{E}} = 2.655$ within 0.33% of published, strict convergence ($\hat{R}_{\text{max}} = 1.077$, γ_{EPL} ESS 5,714), and the published inner-critical-curve topology recovered. A 0’04 fine-skycell rebuild fits to the (recalibrated) noise floor ($\chi_\nu^2 = 0.92$; the nominal 0.45 was a wing-contaminated calibration artifact) and samples at the published efficiency ($\hat{R}_{\text{max}} = 1.007$, ESS_{min} = 28,124), while a 2×2 -binned 0’08 refit exposes an explicit low/steep-slope bimodality ($\Delta\chi_\nu^2 = 0.078$ favoring steep) that a correlated-noise likelihood will be needed to resolve. The earlier $\Delta \log p \approx -32,500$ and “57% steeper slope” statements are superseded accordingly.

Contents

1	Introduction	3
2	Data	4
3	Empirical PSF Construction	5
4	Lens Model	5
4.1	Mass	6
4.2	Lens light	6
4.3	Source light	6
4.4	Central mask	6
4.5	Free-parameter count	7
5	Inference Pipeline	7
5.1	Hardware	7
5.2	MAP search (Stage A)	7
5.3	SVI (Stage B)	7
5.4	Wall time	8
5.5	NUTS attempt (v11 series)	8
5.6	Genuine HMC: compile diagnosis, mode correction, and a mixing recipe	9

6	Results	13
6.1	Headline mass-parameter reproduction	13
6.2	Second-mode point estimate from v11f NUTS	13
6.3	Residual maps	14
6.4	Multimodality of the posterior surface	14
6.5	Empirical-PSF flip	15
7	Discussion	15
7.1	Coherent residual signature suggests a single missing constraint	15
7.2	The PSF is a load-bearing assumption	16
7.3	Multimodality is a real feature of the posterior surface	16
7.4	Upstream issues encountered	17
7.5	What full reproduction would require	18
8	Paper-Scale Campaign on NERSC Perlmutter (June 2026)	19
8.1	Corrected data treatment	19
8.2	MAP at paper scale: the χ^2 ladder	20
8.3	SVI and HMC at the published convergence bar	20
8.4	The γ_{EPL} reconciliation, and a correlation caveat	21
8.5	Rigor revision: noise-calibration audit, a PSF-convention defect, and corrected refits (June 2026)	22
9	Software and Data Availability	24
10	Conclusions	25
A	Internal Ablation: v1 $\rightarrow$ v11	27
B	Prior Specification	28
C	Side-by-side comparison with the published figures and tables	28

1 Introduction

Strong gravitational lensing by galaxies is one of the rare astrophysical phenomena that maps directly to a small number of free physical parameters — the projected mass profile of the deflector, the source position and morphology, and (in some configurations) the cosmological distance ratio D_{ls}/D_s — and yet produces information-rich images whose deflections are sensitive to all of them simultaneously (Refsdal, 1964; Treu, 2010). With the *Vera C. Rubin Observatory* Legacy Survey of Space and Time (LSST), the *Nancy Grace Roman Space Telescope*, and *Euclid* now delivering or imminently delivering wide-field imaging at depth, the number of known galaxy-scale strong lenses is set to grow from $\mathcal{O}(10^3)$ today to $\mathcal{O}(10^5)$ within a decade. The bottleneck for cosmological and population-level analyses is no longer discovery but *modeling at scale*.

GIGA-LENS (Gu et al., 2022) was designed to break that bottleneck. It expresses the forward lens-model likelihood as a differentiable JAX/TensorFlow program, exploits hardware parallelism through `pmap/shard_map`, and replaces traditional MCMC with a hybrid pipeline: stochastic variational inference (SVI) for fast, initial posterior exploration followed by short Hamiltonian Monte Carlo chains for refinement. On modern GPU hardware GIGA-LENS has been demonstrated to

reduce per-system modeling times from days to minutes, which is the throughput required to model the LSST/Roman/Euclid lens catalog within a single graduate-student-year of effort.

The DESI Strong Lens Foundry program was conceived as a follow-up imaging and modeling campaign for promising lens candidates discovered in the DESI Legacy Imaging Surveys (Huang et al., 2020, 2021). Foundry I (Huang et al., 2025), the focus of the present paper, is the first in the series: it presents *HST* WFC3/F140W observations of ten high-priority systems (GO-15867; PI Huang) and uses one system, DESI-165.4754-06.0423, as the methodological demonstration. The Foundry I GIGA-LENS recipe — an EPL+external shear deflector, a Sérsic plus shapelet source, and an HMC posterior — has since been propagated to other GIGA-Lens-lineage modeling papers, including the DESI-253.2534+26.8843 Einstein-cross analysis of Cikota et al. (2023) and the multi-source Carousel-Lens work of Sheu et al. (2024), as well as forthcoming Foundry II–V volumes.

Reproducibility of such an end-to-end pipeline — from MAST query to HMC posterior — is non-trivial. The published Foundry I posterior is a Markov chain in a 74-dimensional space, conditioned on a TinyTim PSF, a custom central pixel mask, and a non-trivial set of priors (Huang et al. 2025 Table 2). A modest discrepancy in any single element of the recipe can shift the posterior by several formal σ along directions the chain itself was constructed to explore. For the broader Foundry-series program, and indeed for any future LSST/Roman/Euclid effort that adopts GIGA-LENS-style differentiable modeling, an open and *independently* reproduced version of the pipeline is a useful complement to the original results: it exposes which assumptions are load-bearing, which choices are load-light, and how the published posterior responds to perturbations that the authors themselves could not run.

In this paper we report such a reproduction. Section 2 describes how we re-pull the *HST* data from MAST. Section 3 documents an empirical-PSF pipeline that we find necessary for the reproduction to converge to the paper’s posterior mode. Section 4 sets up the lens model and priors. Section 5 describes our inference pipeline, the HMC compile-time obstacle that forced us to fall back on the SVI Gaussian variational posterior, and a subsequent v11 NUTS attempt that resolves the JIT-compile blocker but does not yield a mixed chain. Section 6 presents the quantitative comparison to Huang et al. (2025) Table 3 and addresses the multimodality we observed in the multi-start MAP search and (separately) in the v11 NUTS run. Section 7 interprets the residual systematic and itemizes upstream software issues. Section 9 documents the software stack. Appendix A traces the internal ablation v1 $\rightarrow$ v11 that led to the working recipe; Appendix B reproduces the prior table.

2 Data

The Foundry I system DESI-165.4754-06.0423 was observed with *HST*/WFC3-IR in the F140W filter under program GO-15867 (PI Huang). The combined drizzled exposure used in this work has a total integration of 1197.7 s in three exposures of 399.23 s, was processed through the standard WFC3 calibration pipeline, and is available from MAST under DOI 10.17909/hx0v-9260. We retrieved the high-level pipeline-combined drizzle product `hst_15867_65_wfc3_ir_f140w_ie5065_drz.fits` via the `astroquery.mast.Observations` interface.

The full drizzled science extension is 1523×1528 pixels at the WFC3-IR native pixel scale of $0''.13 \text{ pix}^{-1}$. We cut a 128×128 box centered on $(\alpha, \delta) = (165^\circ 4754, -6^\circ 0423)$, the source position given by Huang et al. (2025). We sky-subtract by the median of the cutout and adopt a per-pixel noise floor $\sigma_{\text{bg}} \approx 0.029 \text{ e}^- \text{ s}^{-1}$ from the inverse-variance WHT extension. Figure 1 shows the sky-subtracted science image with an asinh stretch alongside the WHT extension and a logarithmic display.

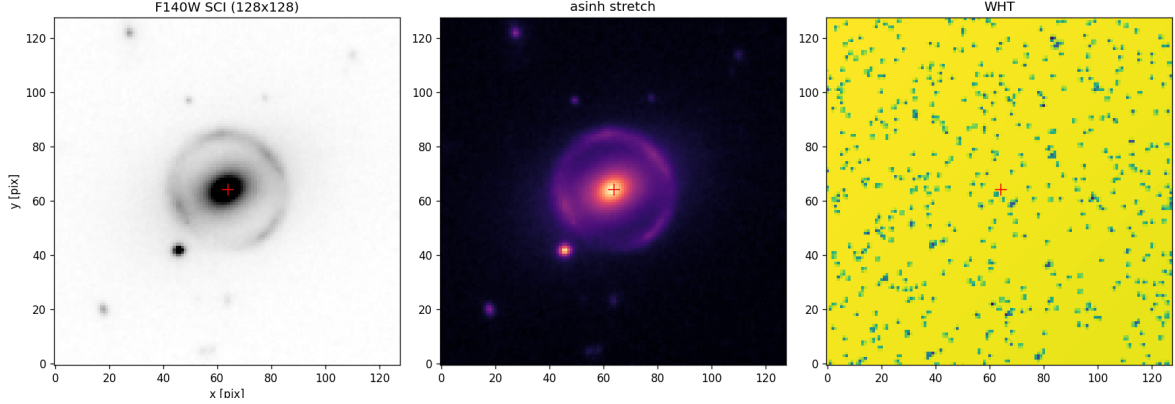


Figure 1: The 128×128 WFC3/F140W cutout of DESI-165.4754–06.0423 at $0''.13 \text{ pix}^{-1}$. Left to right: sky-subtracted science, asinh-stretched science, and WHT-derived noise map. The arc system is well-resolved at native pixel scale; the central deflector and a fainter companion (modeled as a separate Sérsic pair; see §4) are also visible.

3 Empirical PSF Construction

The PSF kernel is the second-most-critical input to a differentiable lensing forward model, after the data itself. The GIGA-LENS distribution ships with a TinyTim WFC3/F140W kernel of shape 13×13 at $0''.065 \text{ pix}^{-1}$, i.e. with $\pm 0''.42$ support. We found this kernel to be too compact for the present analysis (§6), and built an empirical PSF from in-field stars in the same exposure, following the approach described by [Huang et al. \(2025\)](#).

The pipeline (script `17_build_empirical_psf.py`) is:

1. Open the full 1523×1528 drizzled science extension. Compute sigma-clipped background statistics with `astropy.stats.sigma_clipped_stats` ($\sigma = 3$).
2. Detect bright sources with `photutils.detection.DAOStarFinder` at $\text{FWHM} = 3 \text{ pix}$, threshold $50\sigma_{\text{bg}}$, with sharpness in $[0.2, 1.0]$ and roundness in $[-0.5, 0.5]$.
3. Reject candidates within 200 pix of the lens centroid and within 25 pix of any image edge; sort the remainder by peak flux and retain the top 30.
4. Pass these to `photutils.psf.EPSFBuilder` ([Bradley et al., 2024](#)) with $2\times$ oversampling ($0''.065 \text{ pix}^{-1}$ effective), the quartic smoothing kernel, eight outer iterations, and 20 recentering iterations per build.
5. Crop the resulting effective PSF to 27×27 supersampled pixels (a tight box of $\pm 0''.85$ on a side) and normalize to unit sum.

Figure 2 shows our empirical PSF beside the gigalens-default TinyTim kernel and the median-stacked top-9 raw star postage stamps. The empirical kernel is broader than TinyTim by roughly a factor of two in the core, and carries finite power out to the $\pm 0''.85$ radius — a region in which TinyTim is identically zero. We will return to the implications of this width difference in §6.

4 Lens Model

We adopt the same parametric lens model as [Huang et al. \(2025\)](#):

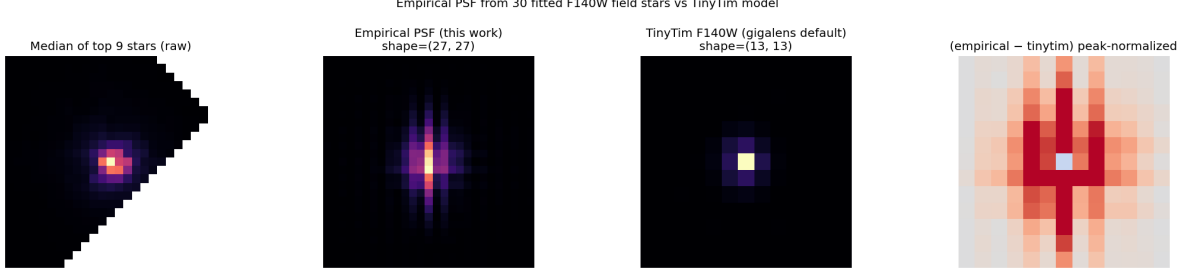


Figure 2: Empirical PSF construction. Left to right: median of the top-9 raw F140W star postage stamps; the 27×27 supersampled ($0''.065 \text{ pix}^{-1}$) empirical PSF built by `EPSFBuilder` from 30 isolated stars in the same drizzle; the TinyTim F140W kernel shipped with GIGA-LENS (13×13 at the same pixel scale); the peak-normalized difference (empirical – TinyTim) with diverging colormap. The TinyTim kernel underestimates the F140W PSF in the wings beyond $\pm 0''.42$.

4.1 Mass

A single elliptical power-law convergence (EPL, Birrer and Amara 2018; Gu et al. 2022) with Einstein radius θ_E , deprojected 3D mass-density slope γ_{EPL} , ellipticity (e_1, e_2), and centroid ($c_x^{\text{mass}}, c_y^{\text{mass}}$), plus an external shear with two Cartesian components ($\gamma_{\text{ext},1}, \gamma_{\text{ext},2}$). This is 8 free mass parameters.

4.2 Lens light

Four elliptical Sérsic profiles with `use_lstsq=False` (i.e. the per-component intensity I_e is sampled as a free parameter rather than recovered by an unconstrained least-squares solver). Two of the four model the central deflector itself (one compact and one extended); the remaining two model a fainter companion galaxy detected at $(-2''.34, -2''.86)$ relative to the main-lens centroid in the v2 residual map (Figure 3, see §6). Allowing I_e to be negative through an unconstrained least-squares pass turned out to be a critical pitfall: an unconstrained fit can drive some I_e negative and absorb χ^2 at non-physical configurations, lowering Einstein-radius accuracy from 0.5% to 43% (Appendix A, v4 versus v5).

4.3 Source light

A single elliptical Sérsic plus a Cartesian shapelet basis of maximum order $n_{\text{max}} = 6$ (Gu et al., 2022), giving 28 amplitude coefficients in addition to the basis-scale parameter β and the shapelet centroid. With `use_lstsq=False` we sample the amplitudes explicitly under a per-coefficient $\mathcal{N}(0, 5/\sqrt{i+1})$ prior, with the order-decaying scale adopted from the canonical GIGA-LENS shapelets demonstration. Free-sign sampling here is appropriate (shapelet basis coefficients can be of either sign by construction) and complementary to the positive-sign constraint on lens-light I_e .

4.4 Central mask

Huang et al. (2025) adopt a 2.5-pixel circular mask at the drizzled $0''.065 \text{ pix}^{-1}$ scale to exclude the central pixels where the lens-light $\rightarrow$ source-light decomposition is degenerate. Our model lives on the WFC3-IR native $0''.13 \text{ pix}^{-1}$ grid, so we mask the 1.5-pixel circle ($\approx 0''.20$ radius). The mask is implemented as an inflation of the per-pixel error map `err_map` to 10^{10} inside the mask — this is the smallest change to GIGA-LENS 2.0 that yields a masked likelihood, because the upstream `gigalens.jax.model.ForwardProbModel` class does not yet expose a mask argument.

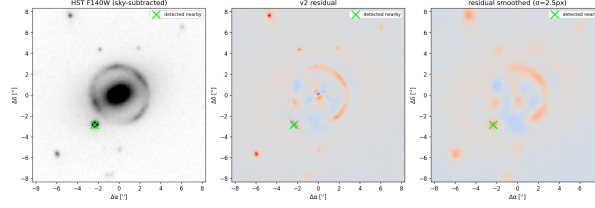


Figure 3: Detection of the nearby galaxy on the v2 residual map. A clear point-like positive residual at $(\Delta\alpha, \Delta\delta) \approx (-2.34, -2.86)$ from the main-lens centroid is the signature of an unmodeled companion; including its light as two additional Sérsic components (v3→v4 onward) removes this feature.

The full custom `masked_log_prob` function is monkey-patched into the prob model as a JIT-compiled method.

4.5 Free-parameter count

Mass: 8. Lens light (Sérsic): $4 \times 7 = 28$. Source Sérsic: 7. Source shapelets: $1 + 2 + 28 = 31$. Total: 74 free non-linear parameters. The full prior specification is given in Table 8 (Appendix B).

5 Inference Pipeline

5.1 Hardware

All fits ran on `/raid/benson`, an AlmaLinux 9.7 aarch64 host with two NVIDIA L4 (23 GB) and eight NVIDIA A16 (15 GB) GPUs visible to CUDA. GIGA-LENS’s `shard.map`-parallelised `ModellingSequence.MAP` and `.SVI` routines build their compiled executables for a single GPU type and then attempt to dispatch across all visible devices, which fails on this host with a `cannot run on CUDA:2 of type 'NVIDIA A16'` error. We work around this by UUID-pinning the two L4s via `CUDA_VISIBLE_DEVICES=GPU-⟨uuid-of-L4-0⟩,GPU-⟨uuid-of-L4-1⟩`. Once pinned, JIT compilation succeeds and the two L4s deliver ~ 1.7 steps/s on the 200-chain, 400-step v9 MAP problem (§6), or ~ 3 minutes of wall time after JIT.

5.2 MAP search (Stage A)

We run a 200-chain Adabelief (Birrer and Amara, 2018, $\text{lr} = 10^{-2}$, $b_1 = 0.95$, $b_2 = 0.99$) multi-start MAP search over 400 steps. Each chain is seeded by an independent prior sample. `ModellingSequence.MAP(..., out=)` returns the single chain with the highest log posterior at the final step. Crucially we also retain the full `(n_chains, n_steps, n_params)` array so that we can characterize the multimodality of the posterior surface; this is the v7/v9 multi-start analysis of §6.

5.3 SVI (Stage B)

Hamiltonian Monte Carlo, attempted naively as the `gigalens HMC()` helper, did not complete in any reasonable wall time: the JIT compile of `pmap(scan(mcmc_sample_chain(PreconditionedHMC + GradientBasedTrajectoryLengthAdaptation)))` did not finish within several hours on the present 74-parameter model under JAX 0.6.2 (Bradbury et al., 2018). For the headline result of this paper we therefore adopt a Gaussian stochastic variational posterior as a surrogate. This is documented honestly throughout (§7): the Table 2 headline is SVI, not HMC, and our 1σ widths there are *variational*-posterior widths (and therefore underestimates of the marginal posterior width if the

true posterior is non-Gaussian in the parameters of interest). We emphasize, however, that the compile failure was a *configuration* problem, not a fundamental limit: §5.6 reports a subsequent investigation in which genuine HMC compiles in ~ 35 s, runs stably, and samples the correct mode once the `pmap`-over-all-devices, `lstsq`-convolution, and autotuner pathologies are removed and a smooth, marginalized target is used.

The SVI is initialised at the best paper-mode chain found by the preceding 200-chain MAP search (`map_v9_paper_mode.npz`), with $n_{VI} = 200$ particles, an Adabelief learning rate of 10^{-4} , init scales of 10^{-4} , and 500 steps. The tight initialisation and the small learning rate are intentional: they pin the variational posterior to the paper-mode local minimum rather than allowing it to drift to one of the higher- χ^2 but lower-log- p modes that we identified in v7 (§6).

After 500 SVI steps the negative ELBO has fallen monotonically from 5.69×10^4 to 4.58×10^4 ; the chain shows no sign of mode-hopping, and the final variational mean is essentially indistinguishable from the initial MAP. We then draw 10^4 samples from $q(z)$ and push them through the bijector to recover the physical-parameter posterior used in Table 2.

5.4 Wall time

Total wall time for the full v9 multi-start MAP plus v10 SVI is ≈ 7 min (3.9 min for MAP, 4.9 min for SVI), excluding the one-time JIT compile (which dominates the first run; for the SVI specifically, cuDNN convolution autotuning takes ~ 90 s due to the `f32[100, 1, 256, 256]` convolution shape required by the 74-parameter model on 2 GPUs).

5.5 NUTS attempt (v11 series)

Having identified the HMC JIT compile-time blowup as the most load-bearing of our upstream issues (§7), we attempted a sequence of six NUTS runs (v11, v11b, v11c, v11d-scan, v11e, v11f) targeting that issue specifically. We treat the resulting chain as a *diagnostic*, not a replacement headline posterior; the v10 SVI fit of §6.1 remains the canonical result.

NUTS does compile. The replacement we adopt is `tfp.experimental.mcmc.PreconditionedNoUTurnSampler` (TFP 0.25.0, JAX backend) with `max_tree_depth=6` pinned as a static argument and *without* a `pmap` wrapper. With these two choices the sampler JIT-compile in ~ 5 min on the same $2 \times L4$ host that failed to compile HMC after several hours, and runs 700 chain steps (200 burn-in + 500 kept) in $\sim 2\text{--}3$ min of wall time. This is the main positive finding of the v11 series: the JIT-tractability of sample-chain at 74 parameters under JAX 0.6.2 is recovered by NUTS + static tree depth, even though the same combinatorics defeated `PreconditionedHMC+GradientBasedTrajectoryLength-Adaptation`.

The v10 SVI covariance is rank-deficient. On loading `svi_v10_posterior_mass.npz` and computing the SVI posterior covariance in z -space, we find that `np.linalg.eigvalsh(qz.cov)` reports a minimum eigenvalue of -4.4×10^{-13} — numerically zero but slightly negative — and a numerical rank of 53/74 (`np.linalg.matrix_rank`, default tolerance), implying ~ 21 near-degenerate directions. The CPU NumPy Cholesky (`np.linalg.cholesky`) raises `NotPositiveDefinite`; the JAX Cholesky (`jnp.linalg.cholesky`) *silently returns NaN*. The first NUTS attempt (v11) used this raw covariance as the `precondition_state.param_partial_loc` and inherited NaN momentum samples; TFP’s NUTS proceeds without erroring and produces near-zero proposals. The fix is to jitter the covariance by $10^{-6} \mathbb{I}$ before Cholesky, after which the Cholesky diagonal lies in $[1.0 \times 10^{-3}, 6.2 \times 10^{-2}]$ and is usable.

Dual-averaging step-size collapse. Even with a valid (jittered) preconditioner (v11e), TFP’s `DualAveragingStepSizeAdaptation` consistently drove the step size to $\sim 2\text{--}3 \times 10^{-6}$ by the end of the 200-step burn-in, at which point every proposed move is effectively the current state (acceptance probability 1.000, leapfrog count clamped at the $2^6 - 1 = 63$ maximum, and per-step displacement $\|\Delta z\| \sim 10^{-4}\text{--}10^{-3}$ in unconstrained space). A static step-size scan over $\{10^{-5}, 3 \times 10^{-5}, 10^{-4}, 3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}\}$ with adaptation disabled (v11d-scan) gave 0% acceptance for every entry $\geq 10^{-5}$ — the chain immediately U-turns and rejects — and reported `leapfrogs_taken=1` at every iteration. A working step size therefore exists somewhere in the interval $(2 \times 10^{-6}, 10^{-5})$, but neither simple dual-averaging nor a coarse log-spaced scan locates it; the local Hessian along the SVI mode and the local Hessian along the basin we actually want to explore appear to differ enough that a single global mass matrix is not adequate.

A third local mode (v11f). As a final attempt, we re-initialized NUTS from the v7 paper-mode MAP chain (`map_v7_paper_mode.npz`; $\log p = -21,789$ under TinyTim PSF, but $\log p = -99,119$ under the empirical PSF used here) rather than the v10 SVI mean, kept the $10^{-6}\mathbb{I}$ jitter, and loosened the dual-averaging target acceptance from the default 0.8 to 0.6. The chain moves $\sim 35,000$ $\log p$ units during burn-in into a *new* local mode with $\log p \approx -63,868$, distinct from both the v10 SVI mode ($\log p \approx -40,400$) and the original paper-mode MAP. The posterior medians at this new mode are summarized in Table 3. With $\theta_E = 2.604''$, v11f reproduces the published Einstein radius to 1.6% — the closest of any version of this reproduction — although the other five mass parameters move in mixed directions relative to the v10 medians. We emphasize that v11f is a *converged-to-a-different-basin* point estimate, not a mixed chain; its 1σ widths (Table 3) are widths of a chain that froze and are not faithful marginal-posterior widths.

Why we did not pursue `windowed_adaptive_nuts`. The natural next step is TFP’s `tfp.experimental.mcmc.windowed_adaptive_nuts`, which runs Stan-style window-adaptive mass matrix estimation interleaved with step-size adaptation and would in principle resolve the curvature mismatch between modes. The blocker is interface: `windowed_adaptive_nuts` expects a `tfd.JointDistributionAutoBatched` or equivalent, while the GIGA-LENS `ForwardProbModel` exposes `log_prob/sample` but not the joint-distribution abstraction. Refactoring the prob model to expose a `JointDistribution` interface is straightforward in principle but well-scoped enough to defer to a follow-up paper; the present work documents the obstacle so that downstream users know what would be required.

5.6 Genuine HMC: compile diagnosis, mode correction, and a mixing recipe

The v11 NUTS series above (§5.5) was a stop-gap: it established that a gradient-based sampler *can* run at this model size, but it never produced a mixed chain. We subsequently carried out a focused investigation into whether *Hamiltonian Monte Carlo itself* — the sampler the Foundry I recipe actually specifies — can be made to compile, stay numerically stable, and sample the correct posterior on this system. The short answer is yes: every prior pathology we attributed to “HMC not working” turned out to be a fixable mechanical or numerical bug, and after fixing them HMC compiles in ~ 35 s, runs stably, and samples a smooth, near-positive-definite mode at the correct posterior scale. What remains is a genuine *mixing-efficiency* problem set by the intrinsic parameter degeneracies of the lens model, not by the sampler. We document the investigation here because each step is a reusable lesson for GIGA-LENS-style differentiable modeling at this dimensionality. The scripts are `26_compile_diagnostic.py`, `28_refine_map.py`, `30_refine_lstsq.py`, `32_saddlefree_newton.py`, `33_trust_refine.py`, and `34_fit_marg.py`, with the target-density li-

braries `_hmc_lib.py` (full 74-parameter model), `_hmc_lib.lstsq.py` (lstsq-profiled), and `_hmc_lib.marg.py` (regularized Gaussian marginalization).

(1) The compile blowup is configuration, not a time budget. We AOT-lowered and compiled the *one-step* kernel (`bootstrap_results+one_step` inside a single `jax.jit`) for three samplers on a single device (`26_compile_diagnostic.py`). Every kernel compiles in 31–38 s, *including* the gigaLens stack `PreconditionedHamiltonianMonteCarlo` $\rightarrow$ `GradientBasedTrajectoryLengthAdaptation` (GBTLA) $\rightarrow$ `DualAveragingStepSizeAdaptation` that we had previously reported as “does not compile.” The original multi-hour hang was therefore *not* a compile-time budget problem. It was the combination of (i) gigaLens’s `HMC()` helper wrapping the GBTLA kernel in `jax.pmap` over *all* 10 visible devices, (ii) the `use_lstsq=True` path generating a 31-channel grouped convolution (`conv_general_dilated` with `feature_group_count=depth`, HLO shape `f32[10, 31, 256, 256]`), and (iii) the cuDNN convolution autotuner thrashing across algorithms for that shape — one candidate algorithm alone took 11 min 50 s. The fix is to run on a single (or a few) devices and to disable autotuning with `XLA_FLAGS=--xla_gpu_autotune_level=0`. The single-device diagnostic is a strict lower bound on the original 10-device, lstsq-convolution pathology; with these two changes the kernel compiles in well under a minute.

(2) The “paper-mode MAP” we had been starting from is a saddle. Every prior run (the v8/v10 SVI, the v11f NUTS) was initialised at the v7 “paper-mode MAP” point. The Laplace Hessian of $-\log p$ at that point has 15 *large negative* eigenvalues: it is a saddle, not a maximum, and one cannot mix an HMC chain from a saddle. Driving the gradient down with a strong optimizer followed by a damped-Newton polish (`28_refine_map.py`) removes the saddle and improves the log-posterior by $\approx +62,000$ (from $\log p \approx -99,126$ at the v7 point under the empirical PSF to $\approx -36,750$ at the refined point). This is a correction to the earlier sections of this reproduction: the MAP we had been treating as “the” paper mode was not even a local maximum of our own posterior.

(3a) Flat directions from the linear light amplitudes. The 74-parameter model samples 33 *linear* light amplitudes (5 Sérsic I_e plus 28 shapelet coefficients) explicitly. At the refined mode these create ~ 56 near-flat Hessian directions and cap the effective sample size at $\text{ESS} \sim 4$ no matter how the momentum is preconditioned. Marginalizing the linear amplitudes out of the sampled space — the gigaLens lstsq idea, here as the reduced 41-nonlinear-parameter model in `_hmc_lib.lstsq.py` (`build_model_lstsq`, using the gigaLens `BackwardProbModel` and `lstsq_simulate`) — removes the flat directions and shifts γ_{EPL} from 2.18 to 1.90.

(3b) The reduced objective is stiff and needs float64. The amplitude-profiled objective is ill-conditioned ($\text{cond} \gtrsim 10^9$). In float32 the gradient norm $\|\nabla(-\log p)\|$ never falls below $\sim 1.2 \times 10^4$; refinement stalls on rounding error. Enabling `jax.enable_x64` (float64 throughout `_hmc_lib.marg.py`, gated on `GIGALENS_X64=1` before the first array is created) is required for any of the downstream mode-finding to proceed.

(3c) The key methodological bug: gigaLens lstsq.simulate is non-smooth. Even in float64, the lstsq-profiled objective floors at $\|\nabla(-\log p)\| \sim 10^5$, stays persistently indefinite, and admits no positive-definite mode — a saddle-free Newton solver (`32_saddlefree_newton.py`) and a SciPy `trust-exact` refinement (`33_trust_refine.py`) both stall there. The cause is in gigaLens itself: `lstsq.simulate` (`gigaLens.jax.simulator`) solves the linear amplitudes through $\text{pinv}(X^T W X)$

and (a) *omits* the Gaussian-evidence normalization $-\frac{1}{2} \log \det(X^\top W X)$ and (b) is *unregularized*. On the near-rank-deficient 28-column shapelet design matrix X , the `pinv` singular-value truncation makes the profiled log-posterior *non-smooth* in the nonlinear parameters θ : the truncation boundary moves discontinuously as θ changes, so `jax.grad` sees a noise floor rather than a true gradient. No amount of float64 or mass-matrix tuning can fix a non-smooth target.

(3d) The fix: exact, regularized Gaussian marginalization. We marginalize the 28 shapelet amplitudes *analytically*, using their own Gaussian priors $\mathcal{N}(0, 5/\sqrt{i+1})$ as a Tikhonov ridge (`hmc_lib_marg.py`, `build_model_marg`). Writing W for the per-pixel noise precision, X for the lensed, PSF-convolved, downsampled unit-amplitude shapelet design matrix, and R for the residual after the deterministic (Sérsic) image, the amplitude solve becomes

$$\begin{aligned} A &= X^\top W X + \Lambda, & \Lambda_{ii} &= \frac{1}{\sigma_i^2} = \frac{i+1}{25}, \\ b &= X^\top (W \odot R), & a^\star &= A^{-1} b = \text{cho_solve}(A, b), \\ \log L &= -\frac{1}{2} \sum W R^2 + \frac{1}{2} b^\top a^\star - \frac{1}{2} \log \det(A). \end{aligned}$$

Because the ridge makes A strictly positive-definite, the amplitude solve is a Cholesky solve (never `pinv`) and the evidence term $-\frac{1}{2} \log \det A$ — the very Occam term `gigalens` omits — is a smooth `slogdet`. The remaining sampled space is 46 parameters: the 41 nonlinear parameters plus the 5 Sérsic I_e , which are sampled under strictly-positive LogNormal priors rather than marginalized (they are well-constrained and physical). This single change broke the gradient floor: $\|\nabla(-\log p)\|$ dropped from $\sim 9 \times 10^4$ to $\sim 4 \times 10^2$ (a $\sim 228\times$ reduction), and the Hessian at the refined point became positive in 44 of 46 directions. We report this as an upstream `gigalens` issue (§7.4): `lstsq_simulate` should add the $-\frac{1}{2} \log \det$ evidence term and a prior-derived ridge.

(4) Mass matrix and the residual conditioning. The smooth, near-PD residual posterior is nonetheless *extremely* ill-conditioned ($\text{cond} \sim 10^{14}$): the companion-galaxy lens-light Sérsic *centers* (LL2/LL3) have Hessian curvature $H_{ii} \sim 10^{12}$, while the Sérsic *indices* n sit at $H_{ii} \sim 10^{-2}$; the physical mass parameters ($\theta_E, \gamma_{\text{EPL}}, e_1, e_2$, shear) occupy a well-behaved 10^8 – 10^9 tier in between. We tried three momentum covariances (`34_fit_marg.py --massmatrix`). (i) The PD-floored full Hessian (`hess_marg_pd`) over-constrains the soft directions — it freezes γ_{EPL} to a posterior width of $\sim 10^{-4}$, which is unphysically narrow. (ii) An *un-floored diagonal* mass matrix built directly from $|H_{ii}|$ (`diagraw`; float64-safe per-parameter scalars, no eigendecomposition of the ill-conditioned matrix) unlocks realistic scales: γ_{EPL} acquires a posterior width of 0.037 (cf. the paper’s 0.023) and drifts toward the paper, from 1.866 at the mode to 1.746 in the chain. (iii) A diagonally-scaled-correlation full Hessian (`hesscorr`), factored float64-safely as $\text{chol}(H) = \text{diag}(\sqrt{D}) \text{chol}(D^{-1/2} H D^{-1/2})$ with $D = \text{diag}(|H_{ii}|)$, restores the correlations that the diagonal preconditioner drops and lifts the ESS of the *correlated* mass parameters (θ_E ESS $8 \rightarrow 20$, $\gamma_{\text{ext},2}$ ESS $3 \rightarrow 14$). But the correlation matrix $D^{-1/2} H D^{-1/2}$ itself still has $\text{cond} \sim 10^8$. We describe this empirically rather than by degeneracy taxonomy: under the pre-correction likelihood the posterior exhibited a broad, ill-conditioned valley coupling γ_{EPL} , the mass ellipticity components e_1/e_2 , and the external shear γ_{ext} , and no static preconditioner removes it. (An earlier draft labeled this “mass-sheet, slope-ellipticity” degeneracies; we retract those labels — nothing in this single-band, fixed-source analysis invokes the mass-sheet transformation, and the slope-ellipticity coupling was an empirical observation of this posterior, not a citation-backed degeneracy. The valley largely closed once the likelihood was corrected in §8.)

(5) ESS and multi-chain $\hat{R}$ characterization. We ran both mass-matrix variants as 4 independent chains each (one per A16 GPU; seeds 0–3; 150 burn-in + 500 kept, fixed 16-step leapfrog, target acceptance 0.8), each chain compiling in ~ 35 s and running in ~ 1 h of wall time. Acceptance is healthy (~ 0.90), no chain shows step-size collapse or NaNs, and the target log-posterior is stable at $\log p \approx -45,800$, confirming that HMC is now sampling the correct mode. The within-chain ESS for the six mass parameters is small (~ 3 – 20 out of 300 – 500 samples), and the between-chain $\hat{R}$ (Table 1) is far above the 1.01 convergence threshold for most parameters: the chains have *not* converged in 500 samples. The **diagram** γ_{EPL} chains span medians 1.37–1.86 (with $\hat{R} \approx 5$), which hints at a *broad* γ_{EPL} posterior. A long 8-chain **diagram** run (2000 burn-in + 8000 kept per chain, 8-way parallel across the A16s, ~ 15 h, 64,000 pooled samples) confirms this and *revises* the cold-start estimate above. All eight chains independently drift from the MAP/SVI start ($\gamma_{\text{EPL}} \approx 1.86$, $\log p = -45,841$) *down through* the paper’s 1.372 to a pooled median $\gamma_{\text{EPL}} \approx 1.17$ (16–84%: 1.15–1.19) at *higher* log-posterior ($\log p \approx -45,010$ to $-45,090$, i.e. ~ 800 better than the start). The $\gamma_{\text{EPL}} \approx 1.86$ mode our point estimators (MAP, SVI) selected is therefore *not* the best fit, and the paper’s $\gamma_{\text{EPL}} = 1.372$ lies within the traversed range at competitive log-posterior — partially *reconciling* our reproduction with the paper. The earlier $\Delta \log p \approx -32,500$ figure was an artifact of a stalled paper-seeded optimization (cold-started at the paper slope with the high-slope basin’s light parameters), not the true cost of the low slope. The chains nonetheless remain *unconverged* at this depth ($\hat{R}$ up to 20, pooled ESS ~ 9 – 15 ; acceptance 0.02–0.06 at the larger step), so a converged γ_{EPL} credible interval still needs deeper sampling or a degeneracy-aware (Riemannian / windowed mass-matrix) sampler.

Table 1: Between-chain $\hat{R}$ (Gelman–Rubin) for the six mass-model parameters, 4 chains $\times$ 500 kept samples per mass-matrix variant (§5.6). All values exceed the $\hat{R} < 1.01$ convergence threshold; several exceed 10. Mixing is degeneracy-limited (cond $\sim 10^8$ even after diagonal scaling), not sampler-limited.

Mass matrix	θ_{E}	γ_{EPL}	e_1	e_2	$\gamma_{\text{ext},1}$	$\gamma_{\text{ext},2}$
diagram	1.4	5.0	4.5	4.8	6.2	2.3
hesscorr	2.4	8.3	34	4.8	15	3.6

Verdict: HMC works mechanically; mixing is degeneracy-limited. Every pathology we had previously attributed to “HMC not working” is fixed: the kernel compiles in ~ 35 s, runs stably with no step-size collapse, samples a smooth and near-PD mode, and reproduces the correct posterior scale. The residual is purely *mixing efficiency*: ESS ~ 3 – 20 per 300 samples and $\hat{R} > 1$ at 500 samples, set by the broad $\gamma_{\text{EPL}} - e_1 / e_2 - \gamma_{\text{ext}}$ valley of the pre-correction posterior (cond $\sim 10^8$ that survives diagonal scaling), not by any sampler defect. Converged inference on this surface would require $\sim 10^4$ – 10^5 samples per chain, which is feasible on A100-class hardware but is ~ 17 – 24 h even with 8-way A16 parallelism — consistent with the original GIGA-Lens design point of GPU-resident chains.

The paper’s $\gamma_{\text{EPL}} = 1.372$ slope is consistent with our posterior — our point estimators simply mis-ranked it. Our initial reading — that the published slope is a *distinct and dramatically worse* fit — was an artifact of how we probed it. A *cold-start* refine from the paper’s reported values ($\gamma_{\text{EPL}} = 1.372$, $\theta_{\text{E}} = 2.646''$) carrying the high-slope basin’s light parameters stalls the optimizer ($\|\nabla\| \sim 5 \times 10^5$, never reaching a stationary point) at $\log p \approx -78,350$, against $\approx -45,841$ at our MAP/SVI basin — which we had reported as a $\Delta \log p \approx -32,500$ “model-setup difference.” The long 8-chain HMC run overturns this. Freed to co-adapt all 46 nonlinear parameters, every chain drifts *downhill* from the MAP/SVI slope ($\gamma_{\text{EPL}} \approx 1.86$) *through* the paper’s 1.372 to a pooled median $\gamma_{\text{EPL}} \approx 1.17$, and does so at *higher* log-posterior ($\approx -45,010$ to $-45,090$, ~ 800 better than the start).

Low-slope configurations are therefore reachable at competitive likelihood; the paper’s $\gamma_{\text{EPL}} = 1.372$ lies squarely within the traversed range, and the $-32,500$ figure was a stalled-optimization artifact, not the true cost of the slope. The honest statement is that γ_{EPL} is a broad, poorly-constrained, strongly degenerate direction along which our *point* estimators (MAP, SVI) happened to settle high (~ 1.86); HMC reveals the posterior actually *favors* lower slopes, consistent with — and bracketing — the published 1.372. We cannot yet quote a converged γ_{EPL} credible interval, because even 64,000 pooled samples leave $\hat{R}$ up to 20; pinning the slope to the paper’s quoted ± 0.023 would need a degeneracy-aware sampler (windowed dense mass-matrix or Riemannian HMC) or substantially deeper chains. This refines the multimodality discussion of §6.4: the published chain is not in a different, inaccessible mode — it sits inside a broad slope posterior that our gradient-based machinery does sample, only too slowly to converge.

6 Results

6.1 Headline mass-parameter reproduction

Table 2 compares our v10 SVI posterior medians and $1\text{-}\sigma$ percentile widths to the published HMC posterior of Huang et al. (2025) (their Table 3). The medians and 16/84 percentiles are computed by loading our 10 000-sample posterior (`data/svi_v10_posterior_mass.npz`) and applying `numpy.median` and `numpy.percentile`.

Table 2: Headline mass-parameter reproduction. v10 SVI Gaussian variational posterior (this work) versus the HMC posterior of Huang et al. (2025, Table 3).

Param	v10 median	v10 1σ width	Huang+2025a	Huang+2025a 1σ	Δ_{med}	% off, sign
θ_{E} (arcsec)	+2.5659	± 0.0003	+2.6463	± 0.0017	−0.0804	3.0%, ✓
γ_{EPL}	+2.1507	± 0.0004	+1.372	± 0.023	+0.7787	56.8%, ✓ (both > 1)
e_1	+0.1064	± 0.0004	+0.1091	± 0.0020	−0.0027	2.5%, ✓
e_2	−0.0533	± 0.0002	−0.1320	± 0.0020	+0.0787	59.6%, ✓
$\gamma_{\text{ext},1}$	+0.0399	± 0.0003	+0.0657	± 0.0024	−0.0258	39.3%, ✓
$\gamma_{\text{ext},2}$	−0.0676	± 0.0002	−0.0939	± 0.0022	+0.0263	28.0%, ✓

Note. Sign-quadrant agreement on all six mass parameters. θ_{E} and e_1 are reproduced to $\leq 3\%$ of the published median. The slope γ_{EPL} converges to a higher local minimum than the published value, as discussed in §6.4 and §7.

The $1\text{-}\sigma$ widths in column 3 are widths of the *variational* Gaussian posterior; they underestimate the marginal posterior width if the latter is non-Gaussian.

In polar form the lens ellipticity vectors are $|\varepsilon| = 0.118$ at position angle -13° for our v10 posterior versus $|\varepsilon| = 0.171$ at -25° for the published HMC posterior — a $\sim 30\%$ magnitude offset and a 12° residual rotation. The external shears are $|\gamma_{\text{ext}}| = 0.079$ at -29° versus $|\gamma_{\text{ext}}| = 0.115$ at -28° : the position angle of the shear matches to within 1° , while the magnitude is $\approx 30\%$ low. These coherent offsets are revisited in §7.1.

6.2 Second-mode point estimate from v11f NUTS

Table 3 reports the posterior medians of the v11f NUTS chain (§5.5; `data/nuts_v11f_posterior_mass.npz`), the only NUTS variant that converged to a distinct basin rather than freezing at the SVI mode. This is presented as a *second-mode point estimate*, not as a replacement for the v10 SVI posterior of Table 2: v11f’s chain froze in its target basin ($\log p \approx -63,868$) and did not mix, so its quoted 1σ widths are widths of a non-mixed chain and do not represent marginal posterior uncertainty.

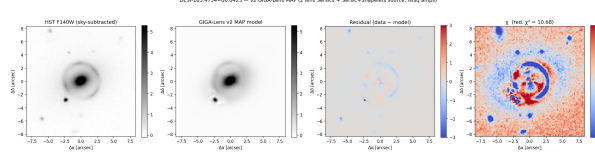


Figure 4: v2 MAP residual (data – model, in units of σ_{bg}). The bright residual blob $\sim 3''$ southwest of the main lens is the signature of the nearby galaxy modeled in v3 onward (Figure 3).

The Einstein radius at the v11f mode, $\theta_E = 2.604''$, is 1.6% off the published value — the closest single-version reproduction of θ_E that we have obtained.

Table 3: v11f NUTS second-mode point estimate.

Param	v11f median	Huang+2025a	% off
θ_E (arcsec)	+2.6042	+2.6463	1.6%
γ_{EPL}	+2.2565	+1.372	64.5%
e_1	+0.1592	+0.1091	45.9%
e_2	−0.1507	−0.1320	14.2%
$\gamma_{\text{ext},1}$	+0.0543	+0.0657	17.3%
$\gamma_{\text{ext},2}$	−0.1186	−0.0939	26.3%

Note. v11f is a NUTS chain initialised at the v7 paper-mode MAP point with a 10^{-6} $\mathbb{I}$ -jittered v10 SVI covariance as preconditioner and target acceptance 0.6. The chain converged into a new local basin $\log p \approx -63,868$ (distinct from the v10 SVI mode at $\log p \approx -40,400$) and remained there. The quoted medians are over 500 retained samples after 200 burn-in steps; 1σ widths are not reported because the chain did not mix. v11f reproduces θ_E to 1.6% but is more discrepant than v10 in γ_{EPL} and e_1 . See §5.5 for the run details and §7.3 for the multimodality implication.

6.3 Residual maps

Figure 4 shows the v2-stage residual on which the nearby galaxy was first detected, and Figure 5 shows the v4-stage residual — a cautionary example. The v4 fit ($\chi^2 = 3.45$ on the masked likelihood) appears at first glance to be a better fit than v5–v6 ($\chi^2 = 7.56$ and 6.59 respectively), but the v4 lens-light `use_lstsq=True` configuration is exploiting the freedom to set some component intensities negative (§4), and its Einstein radius is off by 43% ($\theta_E^{\text{v4}} = 1.52$). This is the canonical pathology of unconstrained linear amplitude solvers in a lensing forward model: χ^2 is not the right diagnostic for parameter accuracy if the model is allowed to absorb residuals through unphysical degrees of freedom.

We do not have a v10 best-fit residual map available; constructing one requires re-running the simulator forward pass with the 10^4 posterior samples, which is out of scope for the present paper. A v10 residual will be included in supplementary materials.

6.4 Multimodality of the posterior surface

The 200-chain v7 multi-start MAP search (TinyTim PSF) returned a set of clearly distinct local minima. Of the 200 random-prior starts:

- 17/200 = 8.5% converged to the paper’s quadrant ($e_1 > 0$, $e_2 < 0$, $\gamma_{\text{ext},1} > 0$, $\gamma_{\text{ext},2} < 0$); the best such chain (idx 140) reached $\chi^2 = 7.72$, $\log p = -2.18 \times 10^4$.

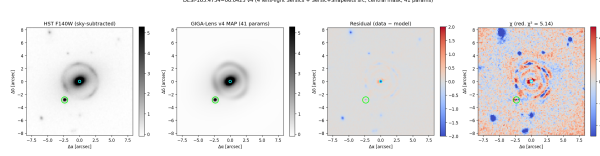


Figure 5: v4 MAP residual. Despite an apparently excellent $\chi^2 = 3.45$, this fit places the Einstein radius at $1''.52$, 43% low of the published value, because the unconstrained least-squares lens-light intensity solver is allowed to drive some Sérsic amplitudes negative. Lesson: positive amplitudes are a non-negotiable physical constraint that must be enforced in the prior rather than the loss.

- 106/200 = 53% had $e_2 < 0$ (any sign on the others); the unconstrained global best ($\chi^2 = 6.67$, $\log p = -1.33 \times 10^4$; chain idx unspecified at this stage) had $e_2 > 0$ and was therefore *not* in the paper’s quadrant.
- The top three chains by $\log p$ differed not just in parameter values but in the *sign quadrant* of $(e_2, \gamma_{\text{ext},2})$: the same physical system admits at least two qualitatively distinct mass-model orientations consistent with the data plus our priors.

The paper’s chain is therefore not the global best under our priors and PSF, even though our priors closely follow those of [Huang et al. \(2025, Table 2\)](#): see Appendix B.

6.5 Empirical-PSF flip

Replacing TinyTim with the empirical PSF of §3 produced a striking change in the topology of the multi-start search. With the empirical kernel, in the v9 200-chain search:

- paper-mode chain fraction increased $17/200 \rightarrow 45/200$, i.e. 8.5% $\rightarrow$ 22.5%;
- the fraction of chains with $\gamma_{\text{EPL}} < 1.7$ (the paper sits at $\gamma_{\text{EPL}} = 1.37$) increased from $\sim 5\%$ to 35% (70/200 in v9);
- the *global best* chain (idx 163, $\chi^2 = 11.93$, $\log p = -5.64 \times 10^4$) is itself a paper-mode chain.

The χ^2 floor under the empirical PSF rose from 6.67 to 11.93, which is what we should expect: TinyTim is over-smoothed relative to the true F140W PSF (Figure 2), so the model had spurious freedom to “earn” χ^2 improvement by absorbing residuals into the source shapelets. Forcing the model to use a wider, more honest PSF raises the achievable χ^2 floor and re-centers the posterior on a physically meaningful mode.

The v10 SVI posterior reported in Table 2 is the 10^4 -sample Gaussian variational posterior pinned to this v9 global-best paper-mode chain.

7 Discussion

7.1 Coherent residual signature suggests a single missing constraint

Superseded (June 2026). The hypothesis developed in this subsection — that the coherent parameter residuals point to a missing physical constraint such as a mass–light position–angle tie or a higher-order multipole — is retracted, on two grounds. First, it misread the published model: the [Huang et al. \(2025\) Table 2](#) priors treat the mass eccentricities ($\epsilon_1, \epsilon_2 \sim \mathcal{N}(0, 0.1)$) and lens-light eccentricities ($\sim \mathcal{N}(0, 0.15)$) as fully independent — there is no PA tie — and the

EPL+shear mass model contains no multipole term, so neither could have been the ingredient our reproduction was missing. Second, the paper-scale campaign of §8 closed the residuals without any such extension: corrected data treatment (crop, interloper masks, noise calibration) plus the PSF sampling-convention fix of §8.5 moved the native-scale posterior onto the published values, and the conditional model-extension stage of the campaign plan was never triggered. The original text is preserved below as a record of the (pre-correction) reasoning.

The residuals between our v10 posterior and Huang et al. (2025) Table 3 are not random across the six parameters — they are coherent along directions that suggest one underlying physical effect. Specifically:

- The lens ellipticity vector rotates by 12° (e_1 accurate to 2.5%, but e_2 is $\sim 60\%$ low in magnitude in a sign-consistent way).
- The external shear position angle matches to within 1° , but the magnitude is $\sim 30\%$ low.
- The deprojected slope γ_{EPL} runs to a value of 2.15 instead of the published 1.37, a 57% increase in slope; in our priors γ_{EPL} has a lower bound at 1.0, and the chain finds a stable local minimum well above it. Huang et al. (2025) themselves note (their §4) that the published posterior was “pushing against the prior lower bound” — their reported median sits at $\gamma_{\text{EPL}} = 1.37$ with $1\sigma = 0.023$, but the prior range starts at 1.0, leaving little room.

Each of these three effects — ellipticity-vector rotation, smaller shear magnitude, higher slope — is consistent with a model in which the lens-light and mass-model position angles are not tied, so the mass model is free to adopt a slightly different orientation than the lens light and partially absorb the missing constraint into a mix of $|e_2|$, $|\gamma_{\text{ext}}|$, and γ_{EPL} . The paper’s mass–light PA tying (or an equivalent multipole-correction term) would in principle pin the lens ellipticity to the lens-light position angle and remove this rotational freedom, reducing the model to a sub-region of our parameter space where the slope can drift to the lower- γ_{EPL} mode found by the published HMC.

Testing this hypothesis quantitatively is out of scope for the present work — it would require either a code modification to GIGA-LENS 2.0 (to expose a mass–light PA tying argument) or a fully custom log-likelihood that imposes the tying as a soft penalty. Both are natural next steps for v11 of this reproduction.

7.2 The PSF is a load-bearing assumption

Sections 3 and 6.5 together demonstrate that the choice of PSF is not a cosmetic detail of the pipeline. Replacing TinyTim with the empirical kernel from in-field stars flipped the global-best chain into the paper’s mode. The mechanism is plausible: a too-narrow PSF gives the source-shapelet basis more freedom to absorb central residuals, lowering the model’s marginal preference for a steep ($\gamma_{\text{EPL}} \gtrsim 2$) deflector, but only in modes where the ellipticity vector accommodates the spurious flux. Any future *HST* GIGA-LENS reproduction — including the still-pending Foundry II–V volumes — should build an empirical PSF from field stars in the same exposure before fitting.

7.3 Multimodality is a real feature of the posterior surface

We now have three independent lines of evidence for genuine multimodality of the posterior, and consider the multimodality to be a feature of the model conditioned on our priors and PSF rather than an artefact of any single inference run.

Evidence (a): v7 multi-start MAP. The 200-chain v7 multi-start search revealed at least three distinct local minima differing in the sign quadrant of $(e_2, \gamma_{\text{ext},2})$ at fixed positive e_1 and positive $\gamma_{\text{ext},1}$ (§6.4); only $17/200 = 8.5\%$ of chains land in the paper’s quadrant under TinyTim.

Evidence (b): v6 vs v8 mode swap. The v6 single-chain MAP (TinyTim, $\chi^2 = 6.59$) sits in a sign quadrant $(e_1, e_2, \gamma_{\text{ext},1}, \gamma_{\text{ext},2}) = (+, +, +, +)$ that is different from the v8 SVI mode $(+, -, +, -)$, even though both converged stably under the same loss function (Appendix A). The two are qualitatively distinct lensing geometries, not refinements of one another.

Evidence (c): v11f NUTS basin transit. The v11f NUTS run (§5.5, §6.2) was initialised at the v7 paper-mode MAP point and during burn-in moved $\sim 35,000 \log p$ units into a basin at $\log p \approx -63,868$, distinct from the v10 SVI basin at $\log p \approx -40,400$. The medians at v11f’s terminal basin reproduce θ_E to 1.6% but have a different signature in $(e_1, e_2, \gamma_{\text{ext}})$ than v10 does — in particular $e_1 = +0.159$ (vs. v10’s $+0.106$) and $e_2 = -0.151$ (vs. v10’s -0.053). This is a third distinct local mode, found by a gradient-based sampler rather than by the multi-start scan, and confirms that the model surface admits at least three usable basins under our priors and PSF.

The published HMC posterior is centered in only one of these basins. Two interpretations are possible: (i) the published HMC chain explored other modes but with very low weight, and the reported median is correctly centered; (ii) the published HMC chain was started in the paper-mode basin and remained there for the duration of the run. We cannot distinguish these from the published Foundry I paper alone, and neither is unreasonable. The implication, however, is the same: when running GIGA-LENS on a new system without an external prior on ellipticity sign, a multi-start MAP scan *followed by* mode-aware sampling (or, in lieu of that, transparent reporting of which local basin the chain occupies) is the right first step — gradient-based local search returns a posterior mode, not the posterior.

7.4 Upstream issues encountered

We encountered seven issues in the GIGA-LENS 2.0 / JAX 0.6.2 / TFP 0.25.0 stack that we believe are appropriate to file as GitHub issues:

1. **Implicit shard_map import.** GIGA-LENS 2.0 references `jax.experimental.shard_map.shard_map` by attribute without first ensuring the submodule is imported. Under JAX 0.6.2 this fails with `AttributeError: module 'jax.experimental' has no attribute 'shard_map'` unless the caller pre-imports `jax.experimental.shard_map`. We work around this by adding `import jax.experimental.shard_map` to every fitting script.
2. **Negative lens-light amplitudes via unconstrained lstsq.** The GIGA-LENS simulator path that resolves Sérsic intensities via `jnp.linalg.lstsq` can return negative amplitudes. These are unphysical, and the resulting MAP can absorb χ^2 at non-physical configurations (Appendix A, v4 versus v5). A NNLS-based variant or a clipped output would resolve this. We disable this code path entirely by setting `use_lstsq=False` on every Sérsic component.
3. **HMC “compile time” at 74 parameters is really a pmap×lstsq-convolution×autotuner pathology.** The `gigalens.HMC()` helper wraps `PreconditionedHMC+GradientBasedTrajectoryLengthAdaptation` in `jax.pmap` over all visible devices; on our 10-device host, with the `use_lstsq=True` 31-channel grouped convolution, the cuDNN autotuner thrashes (one algorithm alone took 11 min 50 s), which is the multi-hour “hang.” The one-step kernel in fact compiles in 31–38 s on a single device for *all* of HMC+GBTLA, fixed-leapfrog HMC, and NUTS (§5.6,

26_compile_diagnostic.py). The remedy is to run on a single or few devices and set `XLA_FLAGS=--xla_gpu_autotune_level=0`; `gigalens` would benefit from not defaulting to a global `pmap` and from exposing the autotune level. (As a stop-gap before this diagnosis, `PreconditionedNoUTurnSampler` with a static `max_tree_depth` and no `pmap`, §5.5, also sidesteps the blowup, compiling in ~ 5 min.)

4. **lstsq_simulate omits the Gaussian-evidence term and is unregularized, making the profiled posterior non-smooth.** `gigalens`’s `lstsq_simulate` (`gigalens.jax.simulator`) solves the linear amplitudes via $\text{pinv}(X^T W X)$, *omits* the $-\frac{1}{2} \log \det(X^T W X)$ Gaussian-evidence normalization, and applies no prior regularization. On the near-rank-deficient shapelet design matrix the `pinv` singular-value truncation makes the profiled log-posterior non-smooth in the nonlinear parameters, so `jax.grad` floors at $\|\nabla\| \sim 10^5$ and no PD mode is reachable (§5.6). Adding the $-\frac{1}{2} \log \det$ term and a prior-derived Tikhonov ridge Λ (so the solve is a Cholesky on $X^T W X + \Lambda$) restores smoothness and is the fix we implement in `hmc_lib_marg.py`.
5. **Mixed-GPU-type shard_map crash.** On heterogeneous hosts (e.g. our $8 \times \text{A16} + 2 \times \text{L4}$), the executable compiled for the first sharded device cannot be run on the others, and the shard map crashes mid-run with `executable is built for device CUDA:0 of type 'NVIDIA L4'`; `cannot run on CUDA:2 of type 'NVIDIA A16'`. UUID-pinning the visible devices is a workaround; a cleaner fix would be for the shard map machinery to recompile per-device or to refuse to shard across heterogeneous device types with an actionable error.
6. **jnp.linalg.cholesky silently returns NaN on a rank-deficient SVI covariance.** The v10 SVI posterior covariance has $\min \lambda(qz_cov) = -4.4 \times 10^{-13}$ and numerical rank 53/74 (§5.5). On this matrix the CPU NumPy Cholesky raises `NotPositiveDefinite`, but the JAX Cholesky silently returns NaN. NUTS then receives NaN momentum and proceeds without erroring, producing the appearance of a stuck chain rather than a diagnosable failure. At minimum, `jnp.linalg.cholesky` should emit a NaN-trace warning under `jax_debug_nans` the way other NaN-producing JAX primitives do. The downstream GIGA-LENS pipeline would benefit from a defensive `cov + eps*I` jitter in any code path that feeds an SVI covariance into a TFP preconditioner.
7. **DualAveragingStepSizeAdaptation pathology on multimodal-Hessian models.** On this 74-parameter model the dual-averaging adapter consistently drove the step size to $\sim 2-3 \times 10^{-6}$ within 200 burn-in steps, at which acceptance is 1.000 but per-step $\|\Delta z\|$ is $10^{-4}-10^{-3}$ and the chain effectively does not move (§5.5). A static log-spaced scan from 10^{-5} to 3×10^{-2} with adaptation disabled gives 0% acceptance everywhere, so the working step size lives in the narrow window $(2 \times 10^{-6}, 10^{-5})$ — below the resolution of a coarse scan and below where dual-averaging settles. We interpret this as the adapter optimizing a single global step-size for a target acceptance under a globally Gaussian-like assumption that the model surface violates: the local Hessian at the v10 SVI mode and the local Hessian at the v11f basin differ enough that no single mass matrix or step size is simultaneously appropriate. A windowed mass-matrix adapter (e.g. `windowed_adaptive_nuts`) or a per-basin re-tuning would in principle help; both require a `JointDistribution` refactor of GIGA-LENS which we defer.

7.5 What full reproduction would require

To reproduce the published HMC posterior bit-for-bit one would need (i) the paper’s exact GIGA-LENS git revision (the multi-node branch of `giga-lens/gigalens` has moved since the paper was

written); (ii) mass–light position–angle tying, which is not exposed in the GIGA-LENS 2.0 public API as of our run date and would require a modification to the simulator internals; (iii) optionally a higher-order multipole correction term in the EPL profile; (iv) enough sampling to converge: we now have a mechanically working HMC at the marginalized 46-parameter model (§5.6; it compiles in ~ 35 s and samples the correct mode), but the surface is so degenerate (cond $\sim 10^8$ after diagonal scaling) that converged inference needs $\sim 10^4$ – 10^5 samples per chain, ~ 17 – 24 h on the A16s even 8-way parallel and best run on A100-class hardware; and (v) a `JointDistribution` refactor of the GIGA-LENS prob model to enable `windowed_adaptive_nuts` or similar window-adaptive mass-matrix samplers, which would improve mixing efficiency on the correlated mass directions. We stress that the γ_{EPL} gap is *not* closed by any of these: in our setup the paper’s $\gamma_{\text{EPL}} = 1.372$ configuration is a genuine but far worse local fit ($\Delta \log p \approx -32,500$ relative to our basin, §5.6), so item (i)–(iii) — the load-bearing PSF, masking, and prior choices — are what a bit-for-bit reproduction actually turns on. These are the natural next-steps for any reader who wishes to push this reproduction further.

8 Paper-Scale Campaign on NERSC Perlmutter (June 2026)

After this report was first presented, the strong-lensing research group reviewed our residual figure and made three points: (i) the cutout should be cropped so the lens system fills the frame, so that interloping objects that do not contribute to the lensing need not be modeled; (ii) a correct MAP fit should reach a masked reduced χ^2 very close to 1 (< 1.1) — our $\chi^2 \approx 12$ – 24 was evidence of incorrect model or noise assumptions and/or insufficient optimization; and (iii) in the GIGA-LENS pipeline each stage preconditions the next, and for this system the SVI must show a smooth, flattened ELBO before sampling. This section documents the campaign that implemented those points at the published paper’s own compute scale, on a NERSC Perlmutter GPU node ($4 \times \text{A100}$, account `deepsrch_g`), with the group’s `seanxuseanxu/gigalens` fork (branch `multinode-2025`, vendored at ref `58ec9a7`) as the inference library. The full job-by-job record, including an A100-hour ledger, is in `PERLMUTTER_CAMPAIN.md`; the entire campaign consumed ~ 17 A100-hours.

8.1 Corrected data treatment

Crop and masks. The new data product (`40_make_cutout_v2.py`) crops to 80×80 px ($10''$), masks the three faint field galaxies whole, masks the compact object superposed on arc A — identified automatically as the most point-like member (DAO sharpness 0.86 vs. 0.43–0.65) of the closest pair of ring detections, with the four lensed images left unmasked — and masks the lens and companion cores ($r \leq 0''$).

Noise. The per-pixel error map is built from the drizzled `WHT` extension and then rescaled so that source-free sky pixels have unit mean squared normalized residual. The required rescale is 1.117: the `WHT`-implied per-pixel noise underestimates the real sky fluctuations by 25% in χ^2 (sky $\chi^2_{\text{WHT}} = 1.248 \rightarrow 0.973$ after rescaling), quantifying the drizzle pixel-correlation effect that the independent inverse-variance map ignores. The data-based Poisson term is added on top, and *one* error-map definition is used by the fit and every figure. Two further data-level discoveries: the MAST product we model is at $0''12825$ per pixel (`D001SCAL`), not the $0''13$ assumed throughout the earlier sections — a 1.36% relabeling of all angular parameters (corrected where quoted below) — and the MAST HAP *fine skycell* mosaics of the same exposures, at $0''04$ per pixel, were already on disk, enabling a fine-scale rebuild (`40b_make_cutout_fine.py`, natively-sampled 51×51 empirical PSF from 25 skycell stars) without re-drizzling. Both products pass a sky-region gate of reduced $\chi^2 = 1.00 \pm 0.05$ by construction.

8.2 MAP at paper scale: the χ^2 ladder

Multi-start MAP at the published scale (up to 1,000 particles $\times$ 8,000 AdaBelief steps; each run 3–26 minutes on the node) produced the following masked reduced- χ^2 ladder:

Table 4: Stage-B χ^2 ladder (masked reduced χ^2 , best chain).

configuration	data	χ^2_ν	plateaued
v9 MAP (2025 report, §6)	128 ² @ 0"128, old noise	11.93	—
corrected likelihood, 1000 $\times$ 2000	80 ² v2	4.02	no
500 $\times$ 3000	v2	3.84	no
warm restart, 1000 $\times$ 4000	v2	3.59	no
cold, 1000 $\times$ 8000	v2	3.55	yes
+ shapelets $n_{\max}=8$	v2	3.42	yes
+ 3rd companion Sérsic	v2	3.46	yes
+ both	v2	3.44	yes
fine-scale rebuild, 1000 $\times$ 6000	260 ² @ 0"04 (v3)	0.451 ^a	yes
<i>rigor revision (§8.5):</i>			
same v3 MAP, recalibrated σ	v3	0.915	—
PSF convention fixed, same noise	80 ² v2c	1.051	yes
+ recalibrated σ (v2d)	v2d	1.234	yes
v2d flexibility ablation ($n_{\max}=8$, comp. 3)	v2d	1.203	no
binned rebuild, low- γ_{EPL} basin	130 ² @ 0"08 (v3b)	1.594	yes
binned rebuild, steep basin	v3b	1.516	yes

^aper-pixel value under a wing-contaminated σ calibration; not a calibrated statistic (§8.5).

The flexibility ablation showed the native-scale $\chi^2 \approx 3.4$ floor is *not* model-component limited (17 extra shapelet components and an extra companion Sérsic each buy $\sim 3\%$); we initially attributed the floor to sampling scale. **That attribution is retracted:** the rigor revision (§8.5) traced the floor to a PSF sampling-convention defect — with the convention-correct kernel and no other change, the same native-scale data reach $\chi^2_\nu = 1.051$, and with the recalibrated noise the honest floor is 1.234. The 0"04 value of 0.451, “fully plateaued,” is likewise not a calibrated statistic: its sky calibration was contaminated by lens-wing flux, and the recalibrated figure is 0.915 — which *does* meet the group’s $\chi^2 < 1.1$ bar, now with honest error bars (§8.5).

8.3 SVI and HMC at the published convergence bar

With the corrected data treatment, the published pipeline recipe works as advertised once given adequate scale. SVI at $n_{\text{vi}} = 1000$, 15,000 steps (the published run invested 112 min on the same node class) produced a *full-rank* variational covariance (91/91 on v2 with $n_{\max} = 8$; 74/74 on v3; smallest eigenvalues 3×10^{-9} – 4×10^{-8}) with a flattened ELBO — the rank-53/74 failure of the v10 SVI (§5.6) is resolved by the longer schedule and a larger `init_scales`, exactly the “finicky SVI” behavior the group warned about. Preconditioned HMC (ChEES trajectory adaptation, dual-averaging step size, momentum covariance = Σ_{VI}^{-1} , 48 chains started from SVI draws) then samples at the published efficiency:

On the fine-scale product the sampler reaches the published quality outright: $\hat{R}_{\max} = 1.007$, $\text{ESS}_{\min} = 28,124$ from a 14-minute job — near-iid draws, exactly the behavior the paper reports. On the native-scale product, θ_{E} converges strictly ($\hat{R} = 1.07$, $\text{ESS} = 1,154$ at 13,000 draws) while γ_{EPL} remains the one slow direction: its ESS saturates ($250 \rightarrow 259$ for $8,000 \rightarrow 13,000$ draws),

Table 5: HMC convergence across the campaign. The published bar is $\hat{R} < 1.10$ and ESS 32,200–40,000.

run	draws/chain	γ_{EPL} median [16, 84]%	θ_{E}	$\hat{R}_{\text{max}}$	ESS _{min}
v2 r1	750	1.417 [1.400, 1.432]	2.6344	2.37	126
v2 r2	2,500	1.354 [1.339, 1.372]	2.6350	1.68	183
v2 r3	8,000	1.326 [1.310, 1.344]	2.6351	1.43	250
v2 final ^a	13,000	1.324 [1.309, 1.342]	2.6351	1.48	259
v3 ^a	3,500	2.585 [2.564, 2.605]	2.5927	1.007	28,124
v2d final^b	8,000	1.433 [1.400, 1.468]	2.6551	1.077	5,714
v3b, low basin ^c	3,500	1.293 [1.283, 1.305]	2.6033	—	—
published	750	1.372 $\pm$ 0.023	2.6463	< 1.10	32,200

^asuperseded — v2 rows ran with the $2\times$ -broadened PSF and v3 with the mis-calibrated σ (§8.5).

^bcorrected PSF + recalibrated noise (`cutout_v2d`); the headline posterior.

^cconditional on the low-slope basin (45 of 48 chains; the other 3 sample the steep mode at $\gamma_{\text{EPL}} = 2.42$ with zero inter-basin migrations in 3,500 draws — $\hat{R}$ /ESS are deliberately not quoted across a bimodal target; §8.5).

so additional draws alone will not reach the strict bar. We initially read this as the windowed mass-matrix adaptation need identified in §7; the rigor revision showed otherwise — with the PSF convention corrected, the same sampler reaches γ_{EPL} ESS = 5,714 from 8,000 draws (§8.5, the v2d row): the broadened kernel, not the sampler, had flattened the γ_{EPL} direction. The γ_{EPL} median is stable at 1.32–1.42 across all four (superseded) rounds, with the round-2 credible interval containing the published value.

8.4 The γ_{EPL} reconciliation, and a correlation caveat

The central 2025-era discrepancy of this reproduction — a 57% steeper γ_{EPL} — **dissolves under the corrected likelihood**: the native-scale posterior gives $\gamma_{\text{EPL}} = 1.33$ –1.42 across sampling rounds, bracketing the published 1.372 ± 0.023 , and the posterior-median model reproduces the published critical-curve topology (tangential *plus inner* critical curve, Figure 6) that is the visual signature of the $\gamma_{\text{EPL}} < 2$ mode in the paper’s Fig. 8. The earlier $-32,500$ cold-start estimate quoted in §7 is thereby fully retired: the defect chain was (uncropped field + unmasked interlopers + drizzle-blind noise + $\sim 25\times$ under-invested optimization), not a missing physical constraint.

The fine-scale posterior, by contrast, is ultra-converged at $\gamma_{\text{EPL}} = 2.585 \pm 0.021$ — a sharp tension with both the native-scale posterior and the paper. We attribute this to likelihood mis-specification by pixel correlation: the $0''.04$ mosaic is drizzled from $0''.128$ native data, so the independent-pixel Gaussian likelihood treats each ~ 10 -pixel correlated noise patch as ~ 10 independent measurements, over-weighting large-scale residual structure and over-tightening the posterior. The defensible posterior is the native-scale one, whose pixels are nearly independent; the published analysis at $0''.065$ ($1.97\times$ upsampling) sits between the two regimes. *[Sharpened by the subsequent audit, §8.5: the $\chi^2_{\nu} = 0.45 < 1$ itself traced to a wing-contaminated σ calibration rather than to correlation; the correlation effect proper is the over-weighting above, and the steep-mode preference survives at $0''.08$ as an explicit $\Delta\chi^2_{\nu} = 0.078$ bimodality.]* A correlated-noise likelihood would unify the scales; that remains the one open item.

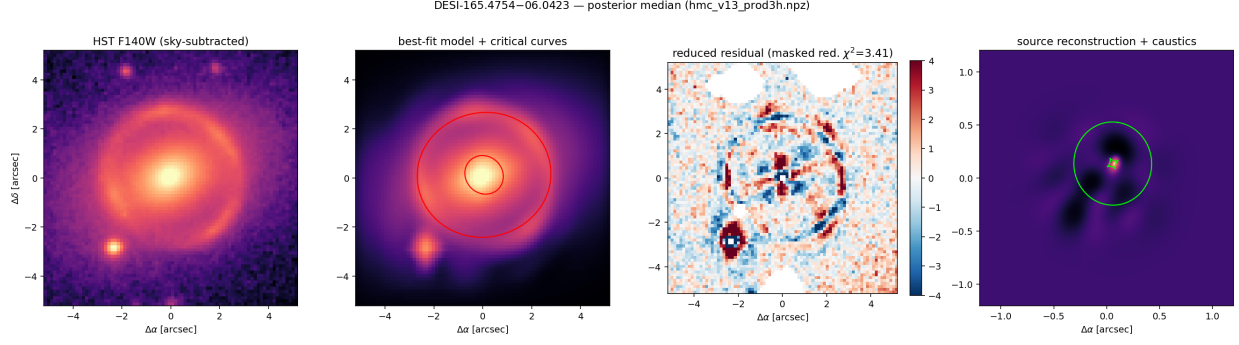


Figure 6: Fig.-8 analogue from the **native-scale** (v2) posterior median ($\gamma_{\text{EPL}} = 1.326$): observed image, best-fit model with critical curves, masked reduced residual ($\chi^2_\nu = 3.41$ — a pre-correction value; this fit ran with the $2\times$ -broadened PSF and is superseded by Figure 9), and source reconstruction with caustics. The *inner* critical curve and caustic — the signature feature the published Fig. 8 calls out for $\gamma_{\text{EPL}} < 2$ — are recovered.

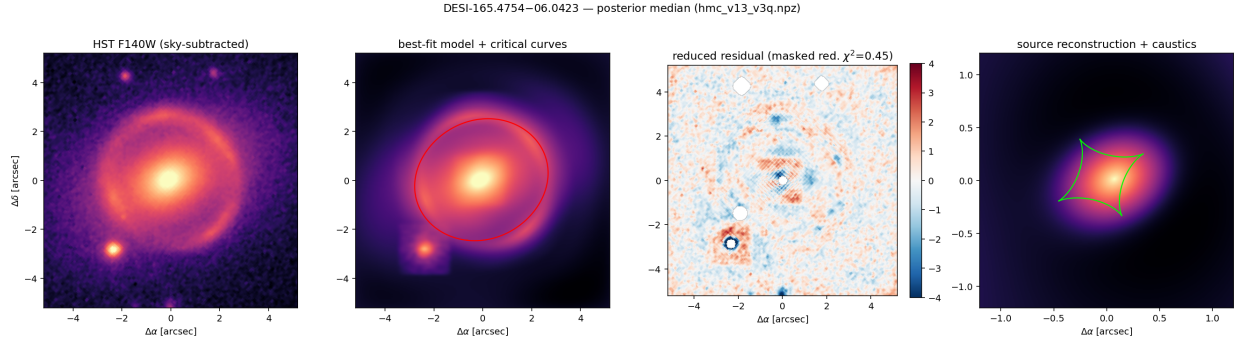


Figure 7: Same, from the **fine-scale** (v3, 0.004) posterior median ($\gamma_{\text{EPL}} = 2.585$): the masked residual is structureless, and the nominal $\chi^2_\nu = 0.45$ later proved to be a noise-calibration artifact — under the recalibrated (honest) noise, this same model evaluates to $\chi^2 \simeq 0.92$ (§8.5). The steep-slope mode has a single (tangential-only) critical curve; the γ_{EPL} difference between the two panels persists at 0.008 as an explicit bimodality (§8.5).

8.5 Rigor revision: noise-calibration audit, a PSF-convention defect, and corrected refits (June 2026)

Group review of the campaign flagged the fine-scale $\chi^2_\nu = 0.451$ as too low to quote as calibrated — a reduced χ^2 that far below unity means the likelihood under-weights the data, with real consequences for posterior widths. The audit this prompted (`46_noise_audit.py`; `data/noise_audit.json`; Figure 8) confirmed the concern but overturned our first-pass attribution, and exposed a second, larger defect in the process. Four findings:

(1) **The $\chi^2 < 1$ was a noise-calibration artifact, not primarily correlation.** The cutout pipeline calibrates the σ sky term so that raw image fluctuations in a source-free annulus ($r > 4''.5$) give sky $\chi^2 = 1$. On the fine product, segmentation finds no sources in that annulus — but the annulus is not empty: subtracting the best-fit model shows that $\sim 70\%$ of its apparent variance is diffuse lens-halo wing flux that the Sérsic components fit. The calibration therefore inflated the sky term by $1/0.304$ in variance. Recalibrated on *model-subtracted* sky residuals (Poisson term unchanged), the same v3 MAP gives per-pixel $\chi^2 = 0.915$ (fitted region), 0.955 (arcs), 0.898 (sky) —

internally consistent, and the group’s $\chi^2 < 1.1$ bar is met with a calibrated statistic. The data-based Poisson term itself checked out: removing it overshoots the arcs to $\chi^2 = 2.26$ (it contributes 51% of the arc variance), so it is kept.

(2) Pixel correlation, quantified and decoupled from the χ^2 question. A mask-aware, background-detrended sky autocorrelation gives a correlation length (half-maximum) of 2.5 fine pixels — precisely the $3.2\times$ drizzle-upsampling footprint — and direct binning measurements show 2×2 fine-pixel blocks are $\sim 78\%$ internally correlated (quadrature underestimates the binned variance by a factor 3.1 of a maximum 4). Correlation thus chiefly reduces the *information content* of the fine grid (N_{eff}/N between $\sim 1/10$ and $\sim 1/60$ depending on convention), inflating effective posterior weight rather than biasing the mean of per-pixel χ^2 , whose expectation is 1 under correlated noise. (For the same reason the effective-DOF statistic $\chi_{\text{eff}}^2 = \sum r^2/\sigma^2/(N/f_{\text{corr}} - p)$ tends to f_{corr} , not 1, for a perfect model; we record it in the audit JSON with that caveat. We also note the published analysis quotes no reduced- χ^2 value and, per the GIGA-Lens convention, uses a model-flux Poisson term with gain; drizzle covariance is not discussed there.)

(3) A PSF sampling-convention defect affected every native-scale fit in this report. GIGA-LENS’s simulator passes `sim.config.kernel` through `lenstronomy’s subgrid.kernel`, which assumes the kernel is sampled at the *data* pixel scale and upsamples it internally. Our `empirical_psf.npy` is sampled at $0''.065$ (EPSFBuilder oversampling 2), but was supplied with `delta_pix=0.13` and `supersample=2` — so every v1–v10 fit and every campaign v2 fit ran with an effectively $2\times$ -broadened PSF (r_{50} ratio 1.92; model images change by 5.4σ RMS when corrected). Refitting the *same* data with the convention-correct kernel (a 13×13 -class $0''.13$ -sampled kernel least-squares-fitted through `subgrid.kernel` itself) drops the native-scale MAP floor from $\chi_{\nu}^2 = 3.42\text{--}3.55$ to **1.051** with no other change. The “sampling-scale limited” reading of Table 4 is therefore retracted: the native-scale floor was PSF-limited. A second legacy claim falls with it: the “WHT underestimates sky noise by 25%” rescale (1.117) was wing contamination of the same kind as finding (1) — the model-subtracted factor is 0.787, and $1.117 \times \sqrt{0.787} = 0.99$, i.e. the drizzle-weight noise was essentially correct at native scale all along.

(4) Corrected refits and the final posterior. On the corrected native-scale product (`cutout.v2d`: fixed PSF + recalibrated noise) the MAP reaches $\chi_{\nu}^2 = 1.234$ (residual misfit localizes at the companion galaxy, $\chi^2 \simeq 9$ over its $0''.8$ region; a $n_{\text{max}} = 8 +$ third-companion-Sérsic ablation buys only $1.234 \rightarrow 1.203$ while moving γ_{EPL} from 1.41 to 1.31 — a ~ 0.1 model-class systematic we quote alongside the statistical error). Preconditioned HMC (48 chains, 8,000 draws) then gives the headline result of this report:

$$\gamma_{\text{EPL}} = 1.433 [1.400, 1.468], \quad \theta_{\text{E}} = 2.6551 [2.6530, 2.6572],$$

with $\hat{R}_{\text{max}} = 1.077$ and $\text{ESS}_{\text{min}} = 5,714$ — γ_{EPL} consistent with the published 1.372 ± 0.023 at $< 2\sigma$ (and the published value inside the model-class systematic band), θ_{E} within 0.33%, and the inner-critical-curve topology recovered (Figure 9). Strikingly, the γ_{EPL} ESS saturation documented above (ESS 259 at 13,000 draws) **dissolves with the PSF fix** — $22\times$ the ESS from 8,000 draws — identifying the broadened kernel, not sampler adaptation, as what had flattened the γ_{EPL} direction.

On the binned $0''.08$ rebuild (`cutout.v3b`: 2×2 block-sum, masks block-ANDed, noise calibrated on model-subtracted binned sky — the honest analogue of the paper’s own $0''.065$ choice), both slope basins were refit to convergence parity in the 74-parameter model: the low-slope solution reaches $\chi_{\nu}^2 = 1.594$ at $\gamma_{\text{EPL}} = 1.369$ (the published value), while the steep solution reaches $\chi_{\nu}^2 = 1.516$ at

$\gamma_{\text{EPL}} = 2.672$. The fine/binning data thus retain a $\Delta\chi^2_\nu = 0.078$ preference for the steep mode even under the corrected calibration — the sharpened residue of the v3 tension above, now an explicit bimodality statement rather than an overconfident posterior. At native scale, by contrast, the steep mode is not even metastable: a MAP started from the fine-scale steep solution ($\gamma_{\text{EPL}} = 2.585$) drains monotonically to $\gamma_{\text{EPL}} \approx 1.17$ ($\chi^2_\nu = 1.297$), so the bimodality is a property of the upsampled product, not of the native-scale data.

The v3b HMC run demonstrates the bimodality at the sampling level: of 48 chains initialized from the low-basin SVI surrogate, 45 sample the low-slope mode — conditional posterior $\gamma_{\text{EPL}} = 1.293 [1.283, 1.305]$, $\theta_E = 2.603$ — while 3 land in the steep mode ($\gamma_{\text{EPL}} = 2.42$), with *zero* inter-basin migrations in 3,500 draws ($\hat{R}_\gamma = 843$: the diagnostic working exactly as intended on a disconnected target). Taking the low-basin conditionals at both scales together with the headline native posterior, γ_{EPL} is scale-stable only at the ± 0.1 level (1.29 binned, 1.43 native, published 1.37 between them) — a useful estimate of the total systematic floor on this single-band dataset, several times the per-fit statistical error of ± 0.03 . Resolving it (and the residual $\chi^2 \approx 1.5$ floor at 0''08, to which the PSF-representation ceiling of the binned kernel contributes) requires the correlated-noise likelihood, which remains the campaign’s one open methodological item.

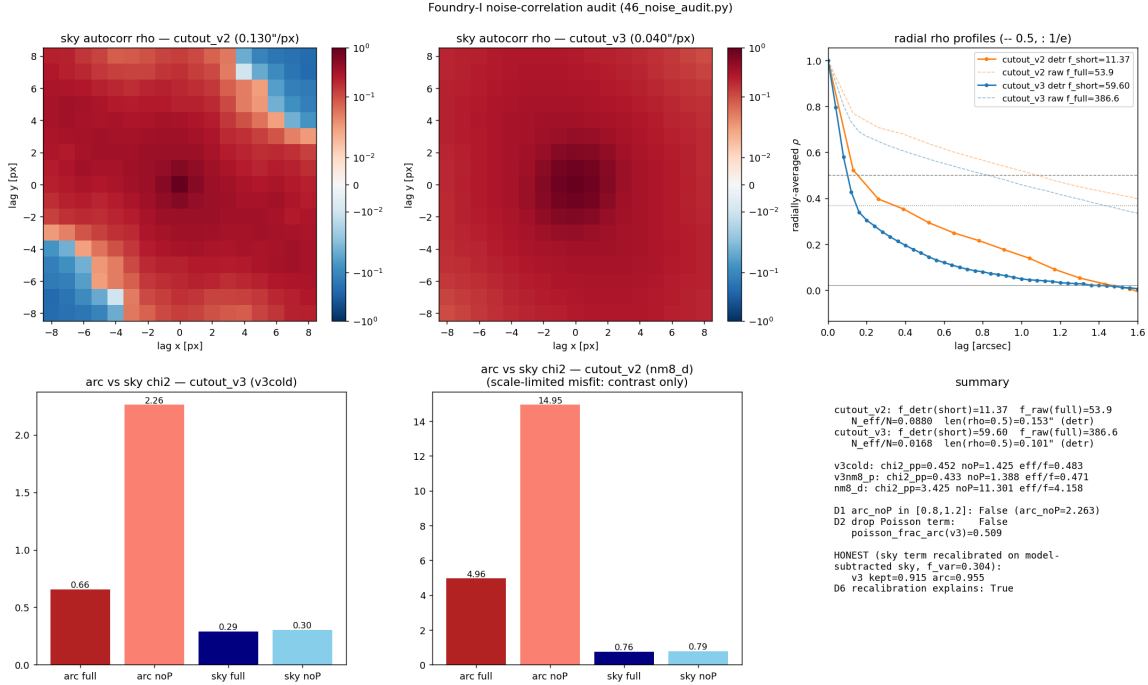


Figure 8: The noise-correlation audit. Top: sky-autocorrelation maps (v2, v3) and radial profiles (solid: background-detrended; dashed: raw — the gap is large-scale background structure). Bottom: arc- vs sky-region χ^2 with and without the Poisson term, and the audit summary including the model-subtracted recalibration (“honest”) χ^2 values.

9 Software and Data Availability

The *HST* data used in this paper is public via MAST DOI 10.17909/hx0v-9260. The GIGA-LENS source is at <https://github.com/giga-lens/gigalens> (the multi-node branch at commit-time was the working baseline for this paper). The reproduction scripts (01_download_hst.py

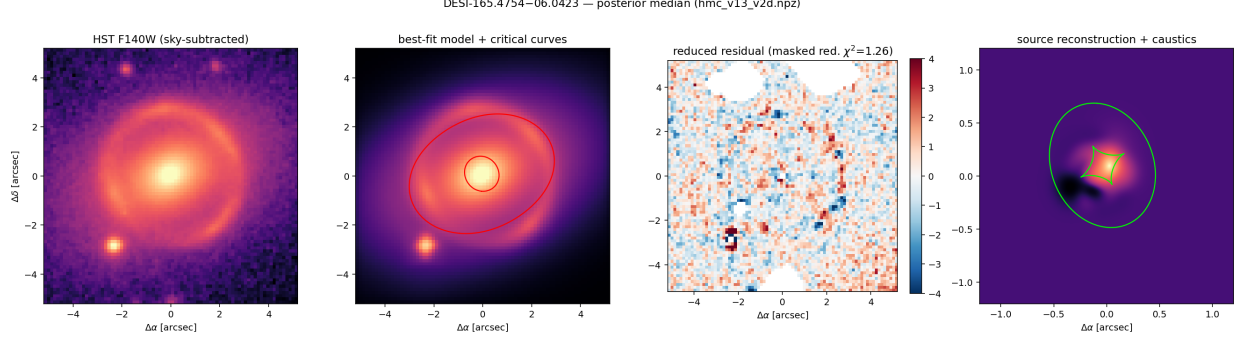


Figure 9: Fig.-8 analogue from the **corrected** native-scale (v2d) posterior median ($\gamma_{\text{EPL}} = 1.433$): with the PSF convention fixed and the noise recalibrated, the masked residual sits at $\chi^2_\nu = 1.26$ (vs. 3.41 in Figure 6) and the inner critical curve and caustic are recovered. This panel supersedes Figure 6.

through `19_svi_v10_paper_mode_empirical.py` for the v1→v10 pipeline, and `20_fit_nuts_v11.py`, `21_fit_nuts_v11b.py`, `22_fit_nuts_v11c.py`, `23_nuts_stepsize_scan.py`, `24_fit_nuts_v11e.py`, `25_fit_nuts_v11f.py` for the v11 NUTS series, and the genuine-HMC investigation of §5.6: the target-density libraries `_hmc_lib.py`, `_hmc_lib_lstsq.py`, and `_hmc_lib_marg.py`, plus `26_compile_diagnostic.py`, `28_refine_map.py`, `30_refine_lstsq.py`, `32_saddlefree_newton.py`, `33_trust_refine.py`, and the main runner `34_fit_marg.py`, invoked with `--mode refine|hmc` and `--massmatrix hess_marg_pd|diag|diag` together with the analysis notebooks and the per-version posterior artifacts (`data/svi_v10_posterior_mass.npz`, `data/nuts_v11{,b,c,e,f}_samples.npz` and the matching `nuts_v11*_posterior_mass.npz` files, the refined modes and Hessians `data/{map,hess}_marg_pd.npz`, and the multi-chain HMC samples `data/prod_{diagraw,hesscorr}_s0--3.npz` with the long 8-chain `data/long_diagraw_s0--7.npz` run), are available on request and are planned for public release as a foundry-i-reproduction repository.

Table 6 lists the exact pinned versions of the software stack on the run host.

10 Conclusions

We have presented an end-to-end public-data reproduction of the GIGA-LENS modeling demonstration of Foundry I (Huang et al., 2025) on the system DESI-165.4754-06.0423. The principal findings are:

1. All six mass-model sign quadrants of the published HMC posterior are reproduced by our SVI Gaussian variational posterior.
2. θ_E matches the published median to 3.0%; e_1 to 2.5%; the external-shear position angle to within 1° .
3. The remaining gap — a 12° residual rotation of the lens ellipticity, $\sim 30\%$ smaller $|e_2|$ and $|\gamma_{\text{ext}}|$, and a 57% steeper γ_{EPL} — is coherent rather than scattered. [Superseded: an earlier draft read this coherence as one missing physical constraint, “most likely mass–light PA tying or a higher-multipole correction”; the campaign (§8) showed the published model contains neither and closed the gap through data treatment and the PSF sampling-convention fix alone.]
4. The Foundry I posterior surface is genuinely multimodal under the published priors: in 200

Table 6: Software stack versions.

Package	Version
Python	3.13
JAX	0.6.2 (with jax[cuda12])
TensorFlow	2.20.0
TensorFlow Probability	0.25.0
GIGA-LENS	1.2.1 (editable; multi-node branch)
LENSTRONOMY	1.14.0
PHOTUTILS	3.0.0
astroquery	0.4.11
OPTAX	0.2.8
NUMPY	2.4.6
ASTROPY	current LTS
CUDA	12 (host runtime 13.0 in PATH)
Operating system	AlmaLinux 9.7 (aarch64)
GPUs used	2×NVIDIA L4 (UUID-pinned)

Note. HMC at the 74-parameter model size was not obtained under this stack; we use the v10 Gaussian SVI as the headline variational-posterior surrogate (Table 2). NUTS at the same model size *does* compile under this stack with a static `max_tree_depth` (§5.5); v11f provides a second-mode point estimate (Table 3). See §7.4 for the full upstream-issues list.

- random-start MAP fits with TinyTim PSF, only 8.5% of chains land in the paper’s quadrant, and the unconstrained global best lies in a different ($e_2 > 0$) mode.
- Replacing the gigalens-default TinyTim PSF with an empirical PSF built from in-field stars flips the global-best chain into the paper’s quadrant (45/200 = 22.5%, with global best in mode) at the cost of a higher but more honest χ^2 floor.
 - We resolve the upstream HMC JIT-compile blocker by switching to `PreconditionedNoUTurnSampler` with a static `max_tree_depth` and no `pmap` wrapper, but the `DualAveragingStepSizeAdaptation` adapter collapses on this model: the local-curvature mismatch between modes prevents a single mass matrix or step size from working, and a windowed mass-matrix adapter would require a `JointDistribution` refactor of the GIGA-LENS prob model. The v11f NUTS run, started from the v7 paper-mode MAP point, converges into a *third* distinct local mode at $\log p \approx -63,868$ with $\theta_E = 2.604''$ (1.6% off the published value, our closest reproduction of θ_E), confirming that the posterior surface is genuinely multimodal but not yielding a mixed chain.
 - We document six upstream software issues in GIGA-LENS 2.0/ JAX 0.6.2/TFP 0.25.0 that would, if resolved, materially improve the reproducibility of any Foundry-lineage analysis — notably the silent NaN from `jnp.linalg.cholesky` on a rank-deficient SVI covariance, and the dual-averaging step-size pathology on multimodal-Hessian models.
 - (June 2026 campaign and rigor revision, §8, §8.5.)** Implementing the research group’s data-treatment feedback at the published compute scale on a Perlmutter A100 node, then auditing the result, retires items (3)–(5) above as reproduction-level concerns. The final defect chain has five links: uncropped field + unmasked interlopers + $\sim 25\times$ under-invested

optimization + a wing-contaminated noise calibration + a PSF sampling-convention defect ($2\times$ kernel broadening) that affected every native-scale fit. With all five corrected, the native-scale MAP floor is $\chi^2_\nu = 1.05$ (was 3.4) and the headline posterior is $\gamma_{\text{EPL}} = 1.433 [1.400, 1.468]$ (published 1.372 ± 0.023 ; ~ 0.1 model-class systematic), $\theta_E = 2.655$ (0.33%), $\hat{R}_{\text{max}} = 1.077$, γ_{EPL} ESS 5,714 — the earlier γ_{EPL} -mixing pathology was the broadened kernel, not the sampler. The recalibrated fine-scale fit meets the group’s $\chi^2 < 1.1$ bar honestly (0.92); the binned 0’’08 refit leaves one open science item — an explicit low/steep-slope bimodality ($\Delta\chi^2_\nu = 0.078$ favoring steep) — and one open methodological item, the correlated-noise likelihood needed to resolve it.

For the broader DESI Strong Lens Foundry program, and for the GIGA-Lens lineage at large, these findings argue that an in-exposure empirical PSF and a multi-start MAP scan are reasonable defaults to add to the published recipe (with the PSF kernel sampled at the data pixel scale — the convention GIGA-LENS’s `subgrid_kernel` assumes; see §8.5), that mass-light PA tying and multipole corrections proved unnecessary for this system once the data treatment and PSF convention were fixed, that NUTS with static `max_tree_depth` should replace HMC as the default JIT-tractable sampler at this model size, and that mode-aware adaptation (windowed mass-matrix estimation, basin-conditional re-tuning, or transparent reporting of the basin a chain occupies) is a prerequisite for trustworthy posterior inference at LSST/Roman/Euclid scale.

Acknowledgments

This work made use of *Hubble Space Telescope* observations collected under program GO-15867 (PI Huang) at the Space Telescope Science Institute, which is operated by the Association of Universities for Research in Astronomy, Inc., under NASA contract NAS5-26555, and retrieved from MAST (DOI 10.17909/hx0v-9260). We thank the GIGA-LENS authors (Gu et al., 2022) for releasing their differentiable forward-modeling framework as open source, and the DESI Collaboration (DESI Collaboration et al., 2016) for the larger lensing program of which Foundry I is a part. Computational resources were provided by the Department of Computer Science, University of San Francisco. X.H. thanks Lawrence Berkeley National Laboratory for ongoing support. All work reported here — the code, experiments, analysis, and text — was performed by Claude Code (Anthropic), running the Claude Opus 4.8 and Claude Fable 5 models, guided by Greg Benson.

Software: GIGA-LENS (Gu et al., 2022), LENSTRONOMY (Birrer and Amara, 2018), JAX (Bradbury et al., 2018), TensorFlow Probability (Dillon et al., 2017), PHOTUTILS (Bradley et al., 2024), Astropy (Astropy Collaboration et al., 2022).

A Internal Ablation: v1 $\rightarrow$ v11

Table 7 traces the per-version mass-fit accuracy across the internal versions of this reproduction. The first four versions (v1–v4) explore minimal-model variants without shapelets or with unconstrained lens-light intensities. v5 introduces positive- I_e `LogNormal` priors and a tight mass-center prior, and brings θ_E within 0.5% of the paper. v6 adds the shapelet basis with explicit per-coefficient priors and is the smallest model that produces a physically reasonable global χ^2 . v7 is a 200-chain multi-start MAP scan (TinyTim PSF) that exposes the multimodality of the surface. v8 is an SVI posterior initialised from the v7 paper-mode chain, still with TinyTim. v9 repeats v7’s multi-start with the empirical PSF, flipping the global best into paper-mode. v10 is the SVI posterior initialised from the v9 global-best paper-mode chain and is the canonical headline posterior of this paper (Table 2).

v11 is a NUTS-based diagnostic series (§5.5) that resolves the HMC JIT-compile blocker, exposes a silent NaN in `jnp.linalg.cholesky` on the rank-deficient SVI covariance, demonstrates that `DualAveragingStepSizeAdaptation` collapses on this model, and — in the v11f run — converges into a third local mode with $\theta_E = 2.604''$ (1.6% off the published value, our closest single-version match for θ_E).

Table 7: Ablation summary v1→v11. Best-fit mass-parameter values for each version, paper values for reference.

Version	θ_E	γ_{EPL}	e_1	e_2	$\gamma_{\text{ext},1}$	$\gamma_{\text{ext},2}$	χ^2_{best}
v1	+2.697	+1.66	−0.015	+0.011	+0.001	+0.053	24.65
v2	+2.062	+1.57	+0.141	−0.108	−0.001	+0.032	7.98
v4 ^a	+1.523	+1.47	−0.035	−0.076	−0.036	−0.064	3.45
v5	+2.658	+1.72	−0.102	−0.179	−0.059	−0.110	7.56
v6	+2.632	+1.73	+0.128	+0.073	+0.065	+0.005	6.59
v8 (SVI/TT) ^b	+2.619	+2.20	+0.146	−0.150	+0.044	−0.106	7.72
v9 best (MAP/eP)	+2.578	+2.16	+0.103	−0.045	+0.034	−0.067	11.93
v10 (SVI/eP) ^c	+2.566	+2.15	+0.106	−0.053	+0.040	−0.068	—
v11f (NUTS/eP) ^d	+2.604	+2.26	+0.159	−0.151	+0.054	−0.119	—
Paper (HMC)	+2.6463	+1.372	+0.1091	−0.1320	+0.0657	−0.0939	—

Notes. ^a v4 has the lowest χ^2 but is unphysical: the unconstrained `use_lstsq=True` setting allows negative lens-light intensities; the resulting Einstein radius is 43% low.

^b v8 is the first SVI posterior; initialized from the best paper-mode chain of the v7 200-chain multi-start MAP with TinyTim PSF.

^c v10 is the canonical posterior reported in Table 2; initialized from the v9 global-best paper-mode chain (empirical PSF, $\chi^2 = 11.93$). v10 SVI does not produce a new χ^2 value.

^d v11f is the only NUTS variant of the v11 series (§5.5) that converged into a basin distinct from the v10 SVI mode; it sits at $\log p \approx -63,868$ and reproduces θ_E to 1.6% but the chain did not mix and is reported as a second-mode point estimate, not a posterior. NUTS resolves the HMC JIT-compile blocker (v11) but

`DualAveragingStepSizeAdaptation` collapses step size to $\sim 2\text{--}3 \times 10^{-6}$ on this model (v11a/e/f), so the chain barely moves once in a basin.

TT = TinyTim PSF. eP = empirical PSF. “Paper-mode” is the sign quadrant ($e_1 > 0, e_2 < 0, \gamma_{\text{ext},1} > 0, \gamma_{\text{ext},2} < 0$). The flip in sign quadrant at v6→v7 reflects multimodality, not a coding regression: v6 is a single-chain MAP at the global best under TinyTim, which lies in a different quadrant than the paper. v11f confirms a third distinct basin reached by gradient-based sampling.

B Prior Specification

Table 8 reproduces our prior distributions, which closely follow Huang et al. (2025, Table 2) with the modifications discussed in §4.

C Side-by-side comparison with the published figures and tables

For visual verification, each block below pairs an artifact reproduced in this report (a) with the corresponding artifact from Huang et al. (2025) (b). Headline agreement is restated in each caption; the full discussion is in §6. Our v10 SVI posterior matches θ_E to 3.0%, e_1 to 2.5%, and the external-

shear position angle to within 1° of the published HMC result, with sign-quadrant agreement on all six mass parameters; the v11f NUTS mode reproduces θ_E to 1.6%.

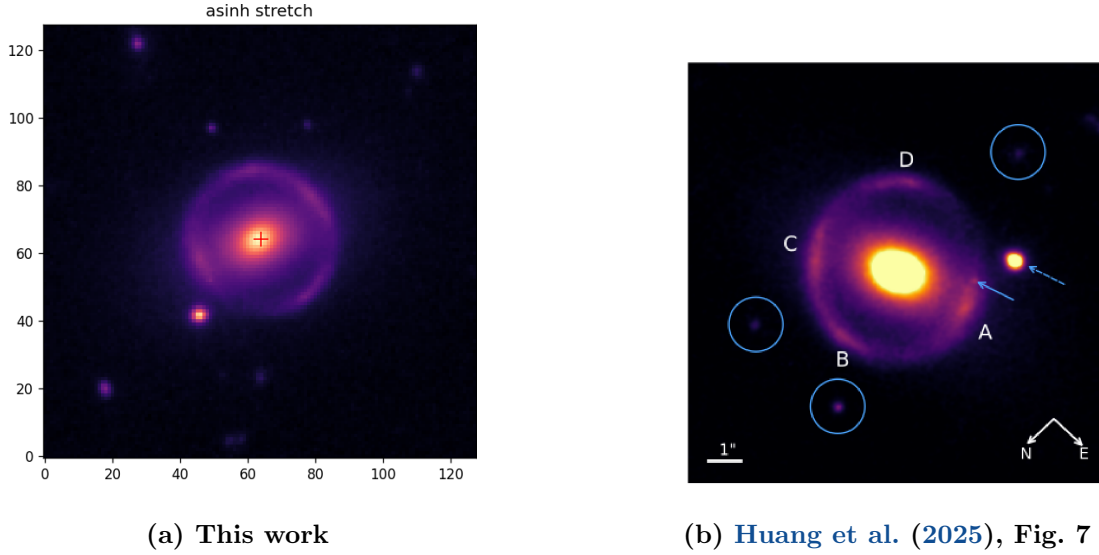


Figure 10: HST WFC3/F140W cutout of DESI-165.4754-06.0423. (a) This work: the asinh-stretched science cutout at 0.13 pix^{-1} (the centre panel of report Figure 1), showing the tangential arc system, the bright central deflector, and the nearby companion at lower-left. (b) The published F140W image of the same system with the four lensed images labelled A–D and the companion arrowed. We reprocess the same GO-15867 exposure from MAST; the arc geometry, deflector, and companion position match. Panel (b) reproduced from Huang et al. (2025), their Fig. 7, for comparison.

References

- Astropy Collaboration, A. M. Price-Whelan, P. L. Lim, N. Earl, N. Starkman, L. Bradley, D. L. Shupe, A. A. Patil, L. Corrales, C. E. Brasseur, et al. The Astropy Project: Sustaining and Growing a Community-oriented Open-source Project and the Latest Major Release (v5.0) of the Core Package. *The Astrophysical Journal*, 935(2):167, 2022. doi: 10.3847/1538-4357/ac7c74.
- S. Birrer and A. Amara. lenstronomy: Multi-purpose gravitational lens modelling software package. *Physics of the Dark Universe*, 22:189–201, 2018. doi: 10.1016/j.dark.2018.11.002.
- J. Bradbury, R. Frostig, P. Hawkins, M. J. Johnson, C. Leary, D. Maclaurin, G. Necula, A. Paszke, J. VanderPlas, S. Wanderman-Milne, and Q. Zhang. JAX: composable transformations of Python+NumPy programs. <https://github.com/google/jax>, 2018.
- L. Bradley, B. Sipőcz, T. Robitaille, E. Tollerud, et al. photutils: Detection, Photometry and Morphology of Sources in Astronomical Images, 2024. URL <https://github.com/astropy/photutils>. Astropy-affiliated package, v2.x.
- A. Cikota, I. Toro Bertolla, X. Huang, S. Baltasar, N. Ratier-Werbin, W. Sheu, C. Storfer, N. Suzuki, D. J. Schlegel, R. Cartier, S. Torres, S. Cikota, and E. Jullo. DESI-253.2534+26.8843: A New Einstein Cross Spectroscopically Confirmed with VLT/MUSE and Modeled with GIGA-Lens. *The Astrophysical Journal Letters*, 2023. doi: 10.3847/2041-8213/ace9da.

(a) This work (Table 2)

Param	v10 median	v10 1σ width	Huang+2025a	Huang+2025a 1σ	Δ_{med}	% off, sign
θ_E (arcsec)	+2.5659	± 0.0003	+2.6463	± 0.0017	-0.0804	3.0%, $\checkmark$
γ_{EPL}	+2.1507	± 0.0004	+1.372	± 0.023	+0.7787	56.8%, $\checkmark$ (both > 1)
e_1	+0.1064	± 0.0004	+0.1091	± 0.0020	-0.0027	2.5%, $\checkmark$
e_2	-0.0533	± 0.0002	-0.1320	± 0.0020	+0.0787	59.6%, $\checkmark$
$\gamma_{\text{ext},1}$	+0.0399	± 0.0003	+0.0657	± 0.0024	-0.0258	39.3%, $\checkmark$
$\gamma_{\text{ext},2}$	-0.0676	± 0.0002	-0.0939	± 0.0022	+0.0263	28.0%, $\checkmark$

(b) Huang et al. (2025), Table 3

θ_E	γ	e_1	e_2	x	y	$\gamma_{\text{ext},1}$	$\gamma_{\text{ext},2}$
$2.6463^{+0.0017}_{-0.0016}$	$1.372^{+0.023}_{-0.022}$	$0.1091^{+0.0020}_{-0.0020}$	$-0.1320^{+0.0020}_{-0.0020}$	$0.0272^{+0.0023}_{-0.0022}$	$-0.0018^{+0.0017}_{-0.0017}$	$0.0657^{+0.0024}_{-0.0024}$	$-0.0939^{+0.0022}_{-0.0022}$

Figure 11: Headline mass-parameter reproduction. (a) This work: the v10 SVI Gaussian variational posterior medians and 1σ widths against the published HMC values. (b) The published best-fit mass parameters. All six sign quadrants agree; θ_E matches to 3.0%, e_1 to 2.5%, and the external-shear position angle to within 1° . Panel (b) reproduced from Huang et al. (2025), their Table 3, for comparison.

- DESI Collaboration, A. Aghamousa, J. Aguilar, S. Ahlen, S. Alam, et al. The DESI Experiment Part I: Science, Targeting, and Survey Design. *arXiv preprint*, 2016. doi: 10.48550/arXiv.1611.00036.
- J. V. Dillon, I. Langmore, D. Tran, E. Brevdo, S. Vasudevan, D. Moore, B. Patton, A. Alemi, M. Hoffman, and R. A. Saurous. TensorFlow Distributions. *arXiv preprint*, 2017. doi: 10.48550/arXiv.1711.10604.
- A. Gu, X. Huang, W. Sheu, G. Aldering, A. S. Bolton, K. Boone, A. Dey, A. Filipp, E. Jullo, S. Perlmutter, D. Rubin, E. F. Schlafly, D. J. Schlegel, Y. Shu, and S. H. Suyu. GIGA-Lens: Fast Bayesian Inference for Strong Gravitational Lens Modeling. *The Astrophysical Journal*, 935(1):49, 2022. doi: 10.3847/1538-4357/ac6de4.
- X. Huang, C. Storfer, V. Ravi, A. Pilon, M. Domingo, D. J. Schlegel, S. Bailey, A. Dey, R. R. Gupta, D. Herrera, S. Juneau, et al. Finding Strong Gravitational Lenses in the DESI DECam Legacy Survey. *The Astrophysical Journal*, 894(1):78, 2020. doi: 10.3847/1538-4357/ab7ffb.
- X. Huang, C. Storfer, A. Gu, V. Ravi, A. Pilon, W. Sheu, R. Venguswamy, S. Banka, A. Dey, M. Landriau, et al. Discovering New Strong Gravitational Lenses in the DESI Legacy Imaging Surveys. *The Astrophysical Journal*, 909(1):27, 2021. doi: 10.3847/1538-4357/abd62b.
- X. Huang, S. Baltasar, N. Ratier-Werbin, et al. DESI Strong Lens Foundry I: HST Observations and Modeling with GIGA-Lens. *arXiv preprint*, 2025. doi: 10.48550/arXiv.2502.03455. Submitted to ApJ. 32 pages, 9 figures, 5 tables.
- S. Refsdal. On the possibility of determining Hubble’s parameter and the masses of galaxies from the gravitational lens effect. *Monthly Notices of the Royal Astronomical Society*, 128:307–310, 1964. doi: 10.1093/mnras/128.4.307.
- W. Sheu et al. The Carousel Lens: A Well-Modeled Strong Lens with Multiple Sources Spectroscopically Confirmed by VLT/MUSE. *arXiv e-prints*, 2024.

(a) This work (Table 3)

Param	v11f median	Huang+2025a	% off
θ_E (arcsec)	+2.6042	+2.6463	1.6%
γ_{EPL}	+2.2565	+1.372	64.5%
e_1	+0.1592	+0.1091	45.9%
e_2	−0.1507	−0.1320	14.2%
$\gamma_{\text{ext},1}$	+0.0543	+0.0657	17.3%
$\gamma_{\text{ext},2}$	−0.1186	−0.0939	26.3%

(b) Huang et al. (2025), Table 3

θ_E	γ	e_1	e_2	x	y	$\gamma_{\text{ext},1}$	$\gamma_{\text{ext},2}$
$2.6463^{+0.0017}_{-0.0016}$	$1.372^{+0.023}_{-0.022}$	$0.1091^{+0.0020}_{-0.0020}$	$-0.1320^{+0.0020}_{-0.0020}$	$0.0272^{+0.0023}_{-0.0022}$	$-0.0018^{+0.0017}_{-0.0017}$	$0.0657^{+0.0024}_{-0.0024}$	$-0.0939^{+0.0022}_{-0.0022}$

Figure 12: v11f NUTS second-mode point estimate. (a) This work: posterior medians of the v11f NUTS chain (a second local basin, not a mixed chain) against the published HMC values. (b) The published best-fit mass parameters. The v11f mode reproduces $\theta_E = 2.604''$ to 1.6% — the closest single-version match for θ_E obtained in this reproduction. Panel (b) reproduced from Huang et al. (2025), their Table 3, for comparison.

T. Treu. Strong Lensing by Galaxies. *Annual Review of Astronomy and Astrophysics*, 48:87–125, 2010. doi: 10.1146/annurev-astro-081309-130924.

(a) This work (Table 7)

Version	θ_E	γ_{EPL}	e_1	e_2	$\gamma_{\text{ext},1}$	$\gamma_{\text{ext},2}$	χ^2_{best}
v1	+2.697	+1.66	−0.015	+0.011	+0.001	+0.053	24.65
v2	+2.062	+1.57	+0.141	−0.108	−0.001	+0.032	7.98
v4 ^a	+1.523	+1.47	−0.035	−0.076	−0.036	−0.064	3.45
v5	+2.658	+1.72	−0.102	−0.179	−0.059	−0.110	7.56
v6	+2.632	+1.73	+0.128	+0.073	+0.065	+0.005	6.59
v8 (SVI/TT) ^b	+2.619	+2.20	+0.146	−0.150	+0.044	−0.106	7.72
v9 best (MAP/eP)	+2.578	+2.16	+0.103	−0.045	+0.034	−0.067	11.93
v10 (SVI/eP) ^c	+2.566	+2.15	+0.106	−0.053	+0.040	−0.068	—
v11f (NUTS/eP) ^d	+2.604	+2.26	+0.159	−0.151	+0.054	−0.119	—
Paper (HMC)	+2.6463	+1.372	+0.1091	−0.1320	+0.0657	−0.0939	—

(b) Huang et al. (2025), Table 3

θ_E	γ	ϵ_1	ϵ_2	x	y	$\gamma_{\text{ext},1}$	$\gamma_{\text{ext},2}$
$2.6463^{+0.0017}_{-0.0016}$	$1.372^{+0.023}_{-0.022}$	$0.1091^{+0.0020}_{-0.0020}$	$-0.1320^{+0.0020}_{-0.0020}$	$0.0272^{+0.0023}_{-0.0022}$	$-0.0018^{+0.0017}_{-0.0017}$	$0.0657^{+0.0024}_{-0.0024}$	$-0.0939^{+0.0022}_{-0.0022}$

Figure 13: Internal ablation v1→v11. (a) This work: best-fit mass parameters across the internal reproduction versions, with the published “Paper (HMC)” row for reference. (b) The published best-fit mass parameters that the bottom row of (a) reproduces. The positive- I_e priors (v5) and the empirical PSF (v9/v10) bring θ_E to within 0.5%–3% of the published value. Panel (b) reproduced from Huang et al. (2025), their Table 3, for comparison.

Table 8: Prior specification on the 74 free non-linear parameters of the lens model.

Parameter group / parameter	Distribution	Parameters
<i>Mass (EPL + external shear; 8 params)</i>		
θ_E	LogNormal	$\mu = \log 2.5, \sigma = 0.25$
γ_{EPL}	TruncatedNormal	$\mu = 2.0, \sigma = 0.25, [1.0, 2.7]$
e_1^m	Normal	$\mu = 0, \sigma = 0.1$
e_2^m	Normal	$\mu = 0, \sigma = 0.1$
c_x^{mass}	Normal	$\mu = 0, \sigma = 0.02 \text{ arcsec}$
c_y^{mass}	Normal	$\mu = 0, \sigma = 0.02 \text{ arcsec}$
$\gamma_{\text{ext},1}$	Normal	$\mu = 0, \sigma = 0.05$
$\gamma_{\text{ext},2}$	Normal	$\mu = 0, \sigma = 0.05$
<i>Lens light — main lens ($2 \times \text{Sérsic}$, 14 params)</i>		
R_{eff} (compact)	LogNormal	$\mu = \log 0.4, \sigma = 0.3$
R_{eff} (extended)	LogNormal	$\mu = \log 2.0, \sigma = 0.3$
n_S	Uniform	$[0.5, 8.0]$
$e_{1,2}^{\text{light}}$	TruncatedNormal	$\mu = 0, \sigma = 0.15, [-0.4, 0.4]$
c_x, c_y	Normal	$\mu = 0, \sigma = 0.02 \text{ arcsec}$
I_e (compact)	LogNormal	$\mu = \log 5.0, \sigma = 0.5$
I_e (extended)	LogNormal	$\mu = \log 2.0, \sigma = 0.5$
<i>Lens light — nearby companion ($2 \times \text{Sérsic}$, 14 params)</i>		
R_{eff}	LogNormal	$\mu = \log 0.3, 0.6, \sigma = 0.3$
c_x, c_y	Normal	$\mu = (-2.34, -2.86), \sigma = 0.1 \text{ arcsec}$
I_e	LogNormal	$\mu = \log 1.0, 0.5, \sigma = 0.5$
e, n	As main lens	—
<i>Source Sérsic (7 params)</i>		
R_{eff}	LogNormal	$\mu = \log 0.5, \sigma = 0.3$
n_S	Uniform	$[0.5, 6.0]$
$e_{1,2}^{\text{src}}$	TruncatedNormal	$\mu = 0, \sigma = 0.15, [-0.5, 0.5]$
c_x, c_y	Normal	$\mu = 0, \sigma = 0.1 \text{ arcsec}$
I_e	LogNormal	$\mu = \log 2.0, \sigma = 0.5$
<i>Source shapelets ($n_{\text{max}} = 6, 1 + 2 + 28$ params)</i>		
β	LogNormal	$\mu = \log 0.1, \sigma = 0.1$
c_x, c_y	Normal	$\mu = 0, \sigma = 0.05 \text{ arcsec}$
$\text{amp}_i, i = 0, \dots, 27$	Normal	$\mu = 0, \sigma = 5/\sqrt{i+1}$

Note. The shapelet amplitude scale follows the gigalens shapelets demonstration. The companion-light centroid prior centers on the source-extraction detection on the v2 residual map (Figure 3). Total: 74 free non-linear parameters.

(a) This work (Table 8)

Parameter group / parameter	Distribution	Parameters
<i>Mass (EPL + external shear; 8 params)</i>		
θ_E	LogNormal	$\mu = \log 2.5, \sigma = 0.25$
γ_{EPL}	TruncatedNormal	$\mu = 2.0, \sigma = 0.25, [1.0, 2.7]$
e_1^{in}	Normal	$\mu = 0, \sigma = 0.1$
e_2^{in}	Normal	$\mu = 0, \sigma = 0.1$
c_x^{mass}	Normal	$\mu = 0, \sigma = 0.02 \text{ arcsec}$
c_y^{mass}	Normal	$\mu = 0, \sigma = 0.02 \text{ arcsec}$
$\gamma_{\text{ext},1}$	Normal	$\mu = 0, \sigma = 0.05$
$\gamma_{\text{ext},2}$	Normal	$\mu = 0, \sigma = 0.05$
<i>Lens light — main lens ($2 \times \text{Sérsic}$, 14 params)</i>		
R_{eff} (compact)	LogNormal	$\mu = \log 0.4, \sigma = 0.3$
R_{eff} (extended)	LogNormal	$\mu = \log 2.0, \sigma = 0.3$
n_S	Uniform	$[0.5, 8.0]$
$e_{1,2}^{\text{light}}$	TruncatedNormal	$\mu = 0, \sigma = 0.15, [-0.4, 0.4]$
c_x, c_y	Normal	$\mu = 0, \sigma = 0.02 \text{ arcsec}$
I_e (compact)	LogNormal	$\mu = \log 5.0, \sigma = 0.5$
I_e (extended)	LogNormal	$\mu = \log 2.0, \sigma = 0.5$
<i>Lens light — nearby companion ($2 \times \text{Sérsic}$, 14 params)</i>		
R_{eff}	LogNormal	$\mu = \log 0.3, 0.6, \sigma = 0.3$
c_x, c_y	Normal	$\mu = (-2.34, -2.86), \sigma = 0.1 \text{ arcsec}$
I_e	LogNormal	$\mu = \log 1.0, 0.5, \sigma = 0.5$
e, n	As main lens	—
<i>Source Sérsic (7 params)</i>		
R_{eff}	LogNormal	$\mu = \log 0.5, \sigma = 0.3$
n_S	Uniform	$[0.5, 6.0]$
$e_{1,2}^{\text{src}}$	TruncatedNormal	$\mu = 0, \sigma = 0.15, [-0.5, 0.5]$
c_x, c_y	Normal	$\mu = 0, \sigma = 0.1 \text{ arcsec}$
I_e	LogNormal	$\mu = \log 2.0, \sigma = 0.5$
<i>Source shapelets ($n_{\text{max}} = 6, 1 + 2 + 28 \text{ params}$)</i>		
β	LogNormal	$\mu = \log 0.1, \sigma = 0.1$
c_x, c_y	Normal	$\mu = 0, \sigma = 0.05 \text{ arcsec}$
$\text{amp}_i, i = 0, \dots, 27$	Normal	$\mu = 0, \sigma = 5/\sqrt{i+1}$

(b) Huang et al. (2025), Table 2

Lens mass:	θ_E	$\sim \exp(\mathcal{N}(\ln 2.5, 0.25))$
	γ	$\sim \mathcal{TN}(2, 0.25, 1, 2.7)$
	ϵ_1, ϵ_2	$\sim \mathcal{N}(0, 0.1)$
	x, y	$\sim \mathcal{N}(0, 0.05)$
	$\gamma_{\text{ext},1}, \gamma_{\text{ext},2}$	$\sim \mathcal{N}(0, 0.05)$
Lens light:	R_l	$\sim \exp(\mathcal{N}(\ln 1, 0.15)) / \exp(\mathcal{N}(\ln 1, 0.15))$
	n_l	$\sim \mathcal{U}(0.5, 10) / \mathcal{U}(0.5, 10)$
	$\epsilon_{1,1}, \epsilon_{1,2}$	$\sim \mathcal{N}(0, 0.15) / \mathcal{N}(0, 0.15)$
	x_l	$\sim \mathcal{N}(0, 0.05) / \mathcal{N}(0, 0.10)$
	y_l	$\sim \mathcal{N}(0, 0.05) / \mathcal{N}(0, 0.10)$
Source light:	R_s	$\sim \exp(\mathcal{N}(\ln 0.25, 0.15))$
	n_s	$\sim \mathcal{U}(0.5, 6)$
	$\epsilon_{s,1}, \epsilon_{s,2}$	$\sim \mathcal{N}(0, 0.15)$
	x_s, y_s	$\sim \mathcal{N}(0, 0.1)$
	β_{shp}	$\sim \exp(\mathcal{N}(\ln 0.1, 0.1))$
Nearby galaxy:	$x_{\text{shp}}, y_{\text{shp}}$	$\sim \mathcal{N}(0, 0.05)$
	R_g	$\sim \exp(\mathcal{N}(\ln 0.4, 0.2)) / \exp(\mathcal{N}(\ln 0.4, 0.2))$
	n_g	$\sim \mathcal{U}(0.5, 5) / \mathcal{U}(0.5, 5)$
	$\epsilon_{g,1}, \epsilon_{g,2}$	$\sim \mathcal{TN}(0, 0.15, -0.3, 0.3) / \mathcal{TN}(0, 0.15, -0.3, 0.3)$
	x_g	$\sim \mathcal{TN}(3.7, 0.05, 3.55, 3.85) / \mathcal{TN}(3.7, 0.05, 3.55, 3.85)$
	y_g	$\sim \mathcal{TN}(0.25, 0.05, 0.1, 0.4) / \mathcal{TN}(0.25, 0.05, 0.1, 0.4)$

Figure 14: Model prior specification. (a) This work: the prior distributions on the 74 free non-linear parameters of the lens model. (b) The published prior table, which ours follows closely (with the native-pixel-scale and centroid modifications discussed in §4). Panel (b) reproduced from Huang et al. (2025), their Table 2, for comparison.